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The World Bank

Report No: 111209

IMPLEMENTATION COMPLETION AND RESULTS REPORT
(IBRD-74060-PY)

ON A

LOAN

IN THE AMOUNT OF US$ 74.0 MILLION

TO THE

REPUBLIC OF PARAGUAY

FOR A

ROAD MAINTENANCE PROJECT

December 30, 2016

Transport and ICT Global Practice
Latin America and Caribbean Region

CURRENCY EQUIVALENTS

(Exchange Rate Effective December 15, 2016)

Currency Unit = Guarani (PYG)
PYG 1.00 = US$0.00017
US$ 1.00 =PYG5,882

FISCAL YEAR
January 1 – December 31

ABBREVIATIONS AND ACRONYMS

CAS	Country Assistance Strategy
CPF	Country Partnership Framework
CREMA	Long Term Rehabilitation and Maintenance Contracts
DNCP	National Directorate of General Procurement
DPV	Road Planning Directorate
DTPC	Transparency and Citizen Participation Directorate
EA	Environmental Assessment
ERR	Estimated Rate of Return
IBRD	International Bank for Reconstruction and Development
IDB	Inter-American Development Bank
IRI	International Roughness Index
GDP	Gross Domestic Product
GMAN's	Performance-based Road Maintenance Contracts
ICR	Implementation Completion Report
IGAP	Government and transparency improvement plan
INDI	National Institute of the Indigenous
IOI	Intermediate Outcome Indicator
LA	Loan Agreement
KPI's	Key Performance Indicators
M&E	Monitoring and Evaluation
MDG	Millennium Development Goals
MOPC	Ministry of public works and communications
NDP	National Development Plan
NPV	Net Present Value
PAD	Project Appraisal Document
PDO	Project Development Objective
PHRD	Japan Policiesand Human Resources Development
PPI	Indigenous People's Plan
RUC	Single Contracting Registration
UPGP	Project Preparation and Management Unit
VMOP	Vice Ministry of Public Works and Communications
VEA	Automatic Reporting System
VOC	Vehicle Operating Cost

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REPUBLIC OF PARAGUAY
Road Maintenance Project

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A. Basic Information			
Country:	Paraguay	Project Name:	Road Maintenance
Project ID:	P082026	L/C/TF Number(s):	IBRD-74060
ICR Date:	12/30/2016	ICR Type:	Core ICR
Lending Instrument:	SIL	Borrower:	REPUBLIC OF PARAGUAY
Original Total Commitment:	USD 74.00M	Disbursed Amount:	US$ 72.0M
Revised Amount:			
Environmental Category: B			
Implementing Agencies: MOPC			
Cofinanciers and Other External Partners: N/A			

B. Key Dates				
Process	**Date**	**Process**	**Original Date**	**Revised / Actual Date(s)**
Concept Review:	11/03/2005	Effectiveness:	01/03/2008	01/03/2008
Appraisal:	06/13/2006	Restructuring(s):		06/06/2012 03/25/2013 01/22/2014 06/16/2015 12/24/2015
Approval:	08/24/2006	Mid-term Review:	09/06/2011	11/29/2011
		Closing:	06/30/2012	06/30/2016

C. Ratings Summary	
C.1 Performance Rating by ICR	
Outcomes:	Moderately Satisfactory
Risk to Development Outcome:	Moderate
Bank Performance:	Moderately Satisfactory
Borrower Performance:	Moderately Satisfactory

C.2 Detailed Ratings of Bank and Borrower Performance (by ICR)			
Bank	**Ratings**	**Borrower**	**Ratings**
Quality at Entry:	Moderately Satisfactory	Government:	Moderately Satisfactory
Quality of Supervision:	Satisfactory	Implementing Agency/Agencies:	Moderately Satisfactory

Overall Bank Performance:	**Moderately Satisfactory**	**Overall Borrower Performance:**	**Moderately Satisfactory**

C.3 Quality at Entry and Implementation Performance Indicators			
Implementation Performance	**Indicators**	**QAG Assessments (if any)**	**Rating**
Potential Problem Project at any time (Yes/No):	Yes	Quality at Entry (QEA):	None
Problem Project at any time (Yes/No):	Yes	Quality of Supervision (QSA):	None
DO rating before Closing/Inactive status:	Moderately Satisfactory		

D. Sector and Theme Codes		
	Original	**Actual**
Sector Code (as % of total Bank financing)		
Central government administration	8	8
Other social services	1	1
Rural and Inter-Urban Roads and Highways	90	90
Sub-national government administration	1	1
Theme Code (as % of total Bank financing)		
Indigenous peoples	17	17
Infrastructure services for private sector development	17	17
Municipal governance and institution building	16	16
Rural services and infrastructure	17	17
Trade facilitation and market access	33	33

E. Bank Staff		
Positions	**At ICR**	**At Approval**
Vice President:	Jorge Familiar Calderon	Pamela Cox
Country Director:	Jesko Hentschel	Axel van Trotsenburg
Practice Manager:	Gylfi Palsson	Jose Luis Irigoyen
Project Team Leader:	Maria Claudia Pachon	Andres G. Pizarro
ICR Team Leader:	Kavita Sethi	
ICR Primary Author:	Stephen Brushett	

F. Results Framework Analysis

Project Development Objectives (from the Loan Agreement)
To establish a sustainable road management strategy that ensures the upgrading and maintenance of the national road network through, *inter alia*,
-the strengthening of the institutional capacity of MOPC,
-the promotion of efficient planning and adequate resource allocation for the rehabilitation and maintenance of the road network, and
- the improvement of access of excluded rural communities to the primary paved road network in selected Departments.

Revised Project Development Objectives (as approved by original approving authority)
Not applicable

(a) PDO Indicator(s)[1]

Indicator	Baseline Value	Original Target Values (from approval documents)	Formally Revised Target Values	Actual Value Achieved at Completion or Target Years
Indicator 1:	**Number of rural community beneficiaries with improved access to all weather roads in the three departments**			
Value (quantitative or qualitative)	0	34,100	n/a	34,100
Date achieved	01/22/2014	06/30/2015	n/a	06/30/2016
Comments (incl. % achievement)	**100%. New indicator** (since 2014). Beneficiaries are defined as those living within 10 km of the improved roads under the Project. According the 2012 census, that number was 33,400. The higher figure of 34,100 represents UPGP's up estimate for mid-2016.			
Indicator 2:	**Increased level of budget execution**			
Value (quantitative or qualitative)	68%	75%	n/a	89%
Date achieved	01/22/2014	06/30/2015	n/a	06/30/2016
Comments (incl. % achievement)	**119%. New indicator** (since 2014). The indicator covers all programs implemented by MOPC, not just those under the Project. The estimate is provided by UPGP based on 2016 performance to date. The estimated level of execution was 83% in 2015.			
Indicator 3:	**Percentage of users reporting satisfaction with the quality of the roads (San Pedro, Caazapa and Caaguazu road segments)**			

Value (quantitative or qualitative)	14% good, 35% fair and 51% bad (from a sample size of 270 road users) (no baseline numbers in 2008)	No target	50% reporting good and 50% reporting fair	77% good, 17% fair and 7% bad (sample size of 637 road users)
Date achieved	01/01/2014	06/30/2012	01/22/2014	06/30/2016
Comments (incl. % achievement)	**Achieved.** Modified the indicator (2014) to only apply to beneficiaries in the three departments where non-paved roads were to be improved. The reported levels of satisfaction are effectively road users' estimates of what is the quality of the road network.			
Indicator 4:	**Road strategy and five year plan implemented**			
Value (quantitative or qualitative)	No strategy	Strategy elaborated and published	n/a	Strategy elaborated and published (approved in December 2006). The five year plan was prepared in 2007 and has been implemented.
Date achieved	01/01/2008	06/30/2012	n/a	06/31/2016
Comments (incl. % achievement)	**Achieved.** The strategy supported an increase in % of paved roads, improvement in the overall condition of the road network. MOPC prepared an updated 5-year plan (2013) and a transport master plan (to 2032)-both helped elaborate the NDP and the Transport Conn. Project.			
Indicator 5:	**Percentage compliance with standards of the length contracted**			
Value (quantitative or qualitative)	0%		90%	93%
Date achieved	01/01/2008	06/30/2012	01/22/2014	06/30/2016
Comments (incl. % achievement)	**103%.** Moved from intermediate to PDO indicator in 2014 restructuring. Is a weighted average of performance of the three GMANS at completion (IBRD funded), i.e.: GMANS 5 - 95%, GMANS 1 - 90% and GMANS 0 - 90%. See 3.2.			
Indicator 6:	**Reduction in vehicle operating costs**			
Value (quantitative or qualitative)	None	None	(see below)	(see below)
Date achieved	01/01/2008	06/30/2012	01/22/2014	06/30/2016

Comments (incl. % achievement)	This PDO indicator was reclassified as an intermediate indicator for component 3 – see below. It is not considered to be a relevant PDO indicator for the Project.			
Indicator 7:	**Reduction in transportation travel times**			
Value (quantitative or qualitative)	None	None	Dropped	n/a
Date achieved	01/01/2008	06/30/2012	01/22/2014	06/30/2016
Comments (incl. % achievement)	This PDO indicator was dropped as part of the January 22, 2014 restructuring. No baseline or target data were provided in the PAD. It is not considered to be a relevant PDO indicator for the Project.			
Indicator 8:	**Number of public transportation vehicles that use the unpaved roads of the (three) targeted departments**			
Value (quantitative or qualitative)	79/km	97/km	42/km	n/a
Date achieved	01/01 /2008	06/30/2012	01/22/2014	06/30/2016
Comments (incl. % achievement)	Reclassified as an intermediate indicator for Component 3. Its assumed that the measure is vehicle-kms not vehicles per km. Not considered relevant PDO indicator for the Project.			

(b) Intermediate Outcome Indicator(s)[2]

Indicator	Baseline Value	Original Target Values (from approval documents)	Formally Revised Target Values	Actual Value Achieved at Completion or Target Years
I: Developing the institutional capacities of the MOPC for strategic planning and management of the road network				
Indicator 1:	**Number of laws, decrees and regulations related to the project enacted**			
Value (quantitative or qualitative)	0	2 (100%)	n/a	5 (250%)
Date achieved	01/01/2008	06/30/2012	n/a	06/30/2016
Comments (incl. % achievement)	**250%.** Transit and Road Safety Law No. 5016 (2014); Road Classif. Law (2016); creation of DPV in MOPC by decree (Feb 2007); toll revenue ministerial directive (rd. maintenance. funding source); and ministerial directive to create transparency department, Feb 2007.			
Indicator 2:	**Creation of Road Strategic Planning Unit**			

Value (quantitative or qualitative)	No unit	Unit operational, equipped and trained	n/a	Unit operational, equipped and trained
Date achieved	01/01/2008	06/30/2012	n/a	06/30/2016
Comments (incl. % achievement)	**Achieved.** The DPV unit is up and running and has adequate funding for equipment and personnel. The enabling measures were taken in February 2007 prior to the declaration of loan effectiveness.			
Indicator 3	**Programming and follow-up of the results-based management and budget implemented**			
Value (quantitative or qualitative)	None	Report on execution of budget for results	n/a	Report on execution of budget for results
Date achieved	01/01/2008	06/30/2012	n/a	06/30/2016
Comments (incl. % achievement)	**Achieved.** A results-based budgeting system for the sector in in place and functional. This provides the basis for regular semi-annual progress reports.			
Indicator 4:	**Strategy of internal and external communication on road maintenance implemented**			
Value (quantitative or qualitative)	No strategy	Reports on % of monitored roads published	Communications plan prepared, implemented and reported. MOPCS performance report prepared, published	Communications plan prepared and implemented (since 2014)
Date achieved	01/01/2008	06/30/2012	n/a	06/30/2016
Comments (incl. % achievement)	**Achieved.** More information being made publicly available (i.e., status of projects under execution), due mainly to the IGAP. Created electronic platform (made it easier to update and enrich base of data and information available for public dissemination).			
II. Promotion of efficient planning and adequate resource allocation for the rehabilitation and maintenance of the road network				
Indicator 5:	**Km of paved network maintained under level of service**			
Value (quantitative or qualitative)	0	960 km through the Project (out of 2,016 km total target)	642 km through Project funding and 629 km supported by IDB	623 km (through Project funding)
Date achieved	01/01/2008	06/30/2012	01/22/2014	06/30/2016

Comments (incl. % achievement)	**97%.** The target was corrected in 2014 to cover roads improved and maintained under the project funding (only 3 out of the original 4 GMANs contracts to be funded by the Project went forward (GMANs 2 was cancelled). See Section 3.2			
Indicator 6:	**Master plan for optimization of roads toll collection developed and implemented**			
Value (quantitative or qualitative)	Outdated toll collection system with 16 stations	Modern integral toll collection system built and equipped with 11 stations	n/a	Modern integral toll collection system built and equipped with 11 stations
Date achieved	01/22/2014	06/30/2012	n/a	06/30/2016
Comments (incl. % achievement)	**Achieved.** Put in place a modern integrated toll system with 11 stations; led to a significant improvement in terms of both user service and revenue generation compared to ex ante status. The works constr. comp. was however subject to cost increases and completion delays.			
Indicator 7:	**Road monitoring system implemented**			
Value (quantitative or qualitative)	No system	Reports of % compliance with strategy	System established	System established
Date achieved	01/01/2008	06/30/2012	n/a	06/30/2016
Comments (incl. % achievement)	**Achieved.** The system was established under the Project, being effectively utilized to measure compliance with the Road Strategy. A system of regular and permanent traffic counting is now in place, starting from 2014.			
III. Improving access for the poorest and most excluded rural communities to the primary paved road network in the Departments of San Pedro, Caaguazu and Caazapa				
Indicator 8:	**Km of the unpaved road network improved**			
Value (quantitative or qualitative)	0	338 km under the Project (out of a total target of 876 km)	140 km	144 km (under Project financing)
Date achieved	01/01/2008	06/30/2012	01/22/2014	06/30/2016
Comments (incl. % achievement)	103%. Target was corrected in 2014 (due to limited funds) to cover only roads to be improved under the project, following the contracting of the GMANs. Improvements included a number of bridges, some works contracts suffered from time and cost overruns.			
Indicator 9:	**Indigenous Peoples Plan prepared and implemented**			
Value (quantitative or qualitative)	No	IPP prepared and implemented with 3 multi-use community centers	n/a	IPP prepared, implemented and started
Date achieved	01/22/2014	06/30/2015	n/a	06/30/2016

Comments (incl. % achievement)	**100%.** Revised the plans in 2011 with only some components initially going forward. Started constr. of multi-use comm. centers late 2015 (difficulties in securing suitable sites and disagreements with INDI). Developed a manual for the operation of the centers 2016.			
Indicator 10	**Maintenance system for unpaved roads is established through micro-enterprises in selected departments**			
Value (quantitative or qualitative)	6	10	n/a	6
Date achieved	01/22/2014	06/30/2015	n/a	06/30/2016
Comments (incl. % achievement)	**Achieved.** Created 6 micro-enterprises by the 2014 restructuring when this indicator was added. However the maintenance system based on the micro-enterprises was successfully established and is ready to be mainstreamed on the unpaved road network.			
Indicator 11:	**Vehicle Operating Cost - Truck: San Pedro, Caazapa, Caaguazu (Guarani per veh-km)**			
Value (quantitative or qualitative)	18149 (San Pedro), 17886 (Caazapa) and 18324 (Caaguazu)	17527 (San Pedro), 17273 (Caazapa) and 17659 (Caaguazu)	10682 (San Pedro), 10497 (Caazapa) and 10569 (Caaguazu)	See Section 2.3.
Date achieved	01/01/2008	06/30/2012	01/22/2014	06/30/2016
Comments (incl. % achievement)	The original target VOC were significantly reduced as part of the January 2014 restructuring based on revised, actual data for 2013 as follows: 11483 (San Pedro), 11283 (Caazapa) and 11360 (Caaguazu).			
Indicator 12:	**Number of Public Transport Vehicles in San Pedro, Caazapá and Caaguazú Road Segments (per km)**			
Value (quantitative or qualitative)	79/km	97/km	42/km	See Section 2.3.
Date achieved	01/01/2008	06/30/2012	01/22/2014	06/30/2016
Comments (incl. % achievement)	The target was reduced during January 2014 restructuring, on the basis of revised date obtained from 2013, putting the baseline at 34 vehicles per km.			
Indicator 13:	**Km of the unpaved road network maintained at appropriate standards**			
Value (quantitative or qualitative)	0	893	n/a	Dropped
Date achieved	01/01/2008	06/30/2012	01/22/2014	06/30/2016
Comments (incl. % achievement)	Indicator dropped in 2014 restructuring. Only 101 km were reported to be under maintenance at the time; Total departmental unpaved roads to be maintained are estimated to be around 1,000km as of close of the Project.			

Indicator 14:	**Number of critical points of difficult transit**			
Value (quantitative or qualitative)	18	0	n/a	Dropped
Date achieved	01/01/2008	06/30/2012	01/22/2014	06/30/2016
Comments (incl. % achievement)	The indicator was dropped as part of the January 22, 2014 restructuring. It is unclear what was the source of the original indicator.			

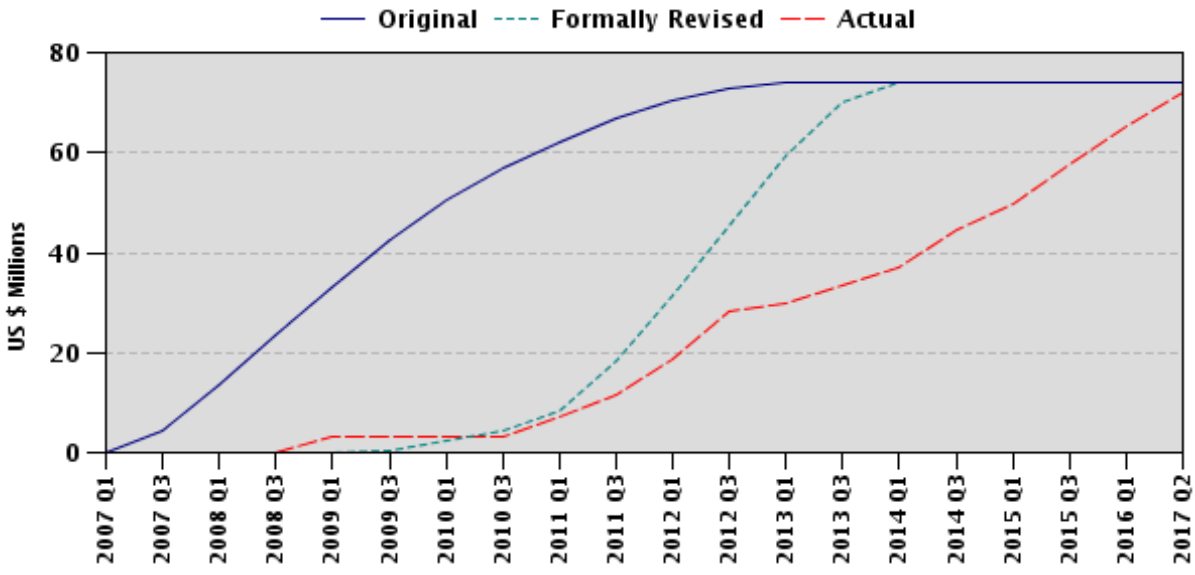
G. Ratings of Project Performance in ISRs

No.	Date ISR Archived	DO	IP	Actual Disbursements (USD millions)
1	11/06/2006	Satisfactory	Satisfactory	0.00
2	05/08/2007	Satisfactory	Satisfactory	0.00
3	06/14/2007	Satisfactory	Moderately Unsatisfactory	0.00
4	12/20/2007	Satisfactory	Moderately Satisfactory	0.00
5	06/25/2008	Satisfactory	Moderately Satisfactory	0.00
6	09/19/2008	Satisfactory	Moderately Satisfactory	3.00
7	12/24/2008	Satisfactory	Moderately Satisfactory	3.00
8	05/15/2009	Satisfactory	Moderately Satisfactory	3.00
9	11/30/2009	Satisfactory	Moderately Satisfactory	3.00
10	02/17/2010	Satisfactory	Moderately Satisfactory	3.00
11	04/21/2010	Satisfactory	Moderately Satisfactory	4.24
12	05/21/2010	Satisfactory	Moderately Satisfactory	4.24
13	11/28/2010	Satisfactory	Moderately Unsatisfactory	7.33
14	02/22/2011	Satisfactory	Moderately Unsatisfactory	10.83
15	08/02/2011	Satisfactory	Moderately Unsatisfactory	13.72
16	02/20/2012	Moderately Unsatisfactory	Moderately Satisfactory	28.14
17	06/20/2012	Moderately Satisfactory	Moderately Satisfactory	28.14
18	07/09/2013	Moderately Unsatisfactory	Moderately Unsatisfactory	35.47
19	11/17/2013	Moderately Satisfactory	Moderately Satisfactory	38.94
20	06/16/2014	Moderately Satisfactory	Moderately Satisfactory	45.83
21	12/30/2014	Moderately Satisfactory	Moderately Unsatisfactory	57.92
22	06/30/2015	Moderately Satisfactory	Moderately Unsatisfactory	60.51
23	12/18/2015	Moderately Satisfactory	Moderately Unsatisfactory	67.31
24	06/30/2016	Moderately Satisfactory	Moderately Unsatisfactory	69.18

H. Restructuring (if any)

Restructuring Date(s)	Board Approved PDO Change	ISR Ratings at Restructuring		Amount Disbursed at Restructuring in USD millions	Reason for Restructuring & Key Changes Made
		DO	IP		
06/06/2012	N	MS	MS	28.14	Extension of the loan closing date to February 28, 2014
03/25/2013	N	MS	MS	35.47	Reallocation of loan proceeds
01/22/2014	N	MS	MSU	44.60	(i) Extension of the loan closing date to June 30, 2015; (ii) Revision of the Results Framework and modification of PDO indicators to better measure PDO.
06/16/2015	N	MS	MU	60.51	Extension of the loan closing date to December 31, 2015
12/24/2015	N	MS	MU	67.31	Extension of the loan closing date to June 30, 2016

I. Disbursement Profile



1. Project Context, Development Objectives and Design

1.1 Context at Appraisal

Country and Sector Background

Paraguay pursued the twin goals of addressing economic stagnation and of fighting corruption. The country's economic program was particularly oriented towards supporting agro-industry and diversification of exports. Addressing governance concerns was a means to help improve both the performance and perception of public institutions. Both sets of goals were seen to require enhancement of human capital, and hence allied plans to increase investment in education and health.

Improved infrastructure was seen as critical for the success of the government's programs. Shortcomings in both the quality and coverage of infrastructure adversely affected the competitiveness of Paraguay's producers, as quality and coverage were seen to be lagging behind regional averages. Specific challenges arose from Paraguay being a land locked country, enhancing the requirement to identify and exploit opportunities to reduce transport and logistics costs. Additionally, better infrastructure, especially in the case of rural roads, were seen as a necessary input to poverty alleviation, through improving access to basic services and business opportunities.

Increased and better focused expenditure on roads was needed to secure quality and coverage improvement. In the road sector more resources were required not only to secure an improvement in the national paved and non-paved networks, but also to sustain maintenance of these networks. At the time of appraisal, the proportion of resources devoted to routine and periodic maintenance was assessed to be far below what was required. The situation was aggravated by a failure of a recently created (and later closed) road fund to provide any incremental resources to maintenance over and above the national budget. Additionally, better focused expenditures needed to be allied with specific measures to improve planning, efficiency and results orientation in the public road agencies as well as to secure greater transparency in decision making.

Road sector policies and institutions were seen as a constraint to improved sector performance. At the time of appraisal, the Government determined a need to improve the ministry responsible for road management, the *Ministerio de Obras Publicas y Comunicaciones* (MOPC). The ministry controlled 10% of the national investment budget and employed 2% of the public sector payroll. MOPC however lacked a clear sector policy oriented primarily towards maintenance, suffered from significant institutional weaknesses, especially those related to planning and operated in an inefficient manner due to gaps in human capacity and poorly adopted internal procedures.

In common with many public sector institutions, MOPC lacked transparency in decision making and suffered from a poor perception form road users and the public in general.

Rationale for Bank Involvement

Road sector investment was seen as highly relevant to the CAS approved in November 2003[3]. The CAS was designed to support financial and fiscal reforms; to help improve governance and transparency in public institutions; to promote sustained growth, especially in rural areas; and to enhance social inclusion in order for Paraguay to meet its Millennium Development Goals (MDG). The proposed lending operation in the road sector was seen to be relevant to both the growth and inclusion dimensions of the CAS. Additionally the operation was expected to facilitate pursuing a transparency agenda specific to the needs of the sector.

The Bank possessed relevant country and sector expertise to help Paraguay improve road sector performance. The Bank had been involved in the road sector in Paraguay over a long period of time, with the most recent operation being the Eighth Highway Project (IBRD Loan 3685-PA). Accordingly, the Bank had a lot of knowledge of the institutional shortcomings of the sector, in particular regard to the planning and carrying out of maintenance, so was well placed to advise on new directions that could be taken to address the issues. The Bank also had by this time developed a lot of experience, globally but more specifically in the Latin America region in countries such as Uruguay and Argentina in the ways to improve road management. This included the increased use of private sector in road maintenance through performance base contracting – an area in which the Government had expressed interest.

Lessons Learned from Past Operations

Present and future focus for support to the sector should be on maintenance. A critical change with respect to past sector operations and priorities was an enhanced focus on maintenance and hence on improving the sustainability of sector investment. The design of the project was to recognize and address specific challenges as follows: establishing planning and expenditure priority for maintenance; addressing weak sector implementation and management; improving budget allocation and reducing the number of competing projects in the sector, through clarifying priorities and targeting a greater share of counterpart funding to maintenance.

Weak sector institutions can hamper implementation performance. With specific reference to the experience of the Eighth Highway Project, insufficient attention given to building and sustaining sector planning and execution capacity can be expected to have a negative effect on sector management and ultimately on project implementation. Thus in addition to strengthening MOPC in its various critical functions, specific actions were

[3] Report No. 27341 dated November 26, 2003

recommended to create a specific focal point for the project, in this case the establishment of the *Unidad de Preparacion y Gestion de Proyecto* (UPGP).

1.2 Original Project Development Objectives (PDO) and Key Indicators)

As stated in the Loan Agreement (LA), the PDO is: **to establish a sustainable road management strategy that ensures the upgrading and maintenance of the national road network through, *inter alia,* the strengthening of the institutional capacity of MOPC, the promotion of efficient planning and adequate resource allocation for the rehabilitation and maintenance of the road network, and the improvement of access of excluded rural communities to the primary paved road network in selected Departments**.

For the purposes of the ICR "to establish a sustainable road management strategy" is considered to be the PDO; its result would be the upgrading and maintenance of the national road network as measured by the three intermediate objectives are those following the word "through ….".

As stated in the PAD, the PDO was to: ***establish a sustainable road management strategy that ensures upgrading and maintenance of the road network through a rational and balanced use of scarce resources.***

The guiding principles were to be improved road conservation within a comprehensive vision that ensures the sustainability of road investments and the preservation of existing road assets. The project's specific objectives were:
(a) developing the institutional capacities of the MOPC for strategic planning and management of the road network;
(b) arresting the deterioration of Paraguay's priority road network, composed of international and regional road corridors, and maintaining its level of service as perceived by road users; and
(c) improving access for the poorest and most excluded rural communities to the primary paved road network in the Departments of San Pedro, Caaguazu and Caazapa[4]

It is noted that the PDO statement in the LA differs from that provided in the PAD in the following respects: firstly, the first specific (or intermediate) objective has been

[4] The PDO as presented in the Annex however has a different wording in the overarching definition as follows: **Improvement of Road Service:** Basic transport infrastructure is maintained in a condition that facilitates the movement of freight and passengers at a cost efficient level of service by arresting further deterioration due to fiscal restrictions and ensuring preservation of existing infrastructure assets in the long term through the rationalization of public expenditures of the road transport sector.

effectively split into two parts, separating "strengthening" from "promotion of efficient planning and adequate resource allocation"; and secondly there is no mention "arresting the deterioration" and "maintaining the level of service" of the road network.

The key indicators (KPIs) selected at Project appraisal to measure the PDO were as follows:

(i) Percentage of MOPC's portfolio executed following results-based management practices;
(ii) Creation of MOPC's Road Strategic Planning Unit and implementation of the Road Strategy;
(iii) Levels of users' satisfaction;
(iv) Average costs of vehicle operation;
(v) Average transportation times; and
(vi) Number of public transportation vehicles that use the unpaved roads of the (three) targeted departments.[5]

1.3 Revised PDO (as approved by original approving authority) and Key Indicators, and reasons/justification

There have been no revisions to the PDO. However the KPIs and the some of the related target values were revised significantly in January 2014 following a corrective restructuring approved by Bank management. There was no impact on expected project outcomes however. Table 1 below compares the original KPI and the revised KPI (see also Section 2.3)

Table 1: Restructuring of Key Performance Indicators (KPIs)[6]

Original KPI in the PAD Text	Baseline	Target Value	New KPI as of January 2014	Baseline	Target Value
Creation of the Road Strategic Planning Unit and Road Strategy and five-year plan implemented (not included in PAD Annex)	No unit, strategy or plan	Unit operational, equipped and trained. **Strategy elaborated and published / Monitoring of Physical and**	***Road Strategy and Five Year Plan implemented*** (creation of a Road Strategic planning Unit becomes an intermediate indicator)..	**No change**	**No change**

[5] The presentation of the KPIs in the PAD text is not identical to the presentation in the Annex 3 where only (iii) though (iv) are considered PDO indicators; (i) does not figure at all and (ii) is considered as two separate intermediate indicators.

[6] Bolded KPIs, baselines and targets are those used in the evaluation of outcome in Section 3

		Financial Execution of the Road Strategy			
Percentage of MOPC's portfolio executed following results-based management Practice (not included in PAD Annex)	No baseline	No target	N/A, though "Programming and follow up of results-based management and budget implemented" was an intermediate indicator for Intermediate Outcome 1[7]	Non-existent	Report on execution of budget results as an intermediate indicator target
			New: Increased level of budget execution	**68%**	**75%**
Increased level of road user satisfaction	No baseline	No target	Retitled as "***Percentage of road users reporting satisfaction with the quality of roads in San Pedro, Caaguazu and Cazaapa***"	**51% bad, 35% fair and 14% good**	**50% reporting good, 50% reporting bad**
Reduction in average vehicle operating costs(truck in PYG per km-veh)	No baseline	No target	Retitled as "Vehicle operating cost – truck San Pedro, Caazapa and Caaguazu in PYG per km – veh" and becomes an indicator for Intermediate Outcome 3 "	11482 (San Pedro), 11283 (Caazapa) and 11360 (Caaguazu)	10682 (San Pedro), 10497 (Caazapa) and 10569 (Caaguazu in PYG per km
Reduction in average transportation times (km/h)	No baseline	No target	Dropped (no data available)	n/a	n/a
			Changed from an indicator for Intermediate Outcome 2 to a PDO indicator: ***Percentage of compliance with standards of the***	**95%**	**90%**

[7] It also appears as such in Annex 3 of the PAD

			length contracted. (%)		
Number of public transportation vehicles that use the unpaved roads of the (three) targeted departments (veh-km)	79 per vehicle-km	97 per vehicle-km	Becomes an indicator for Intermediate Outcome 3 and is retitled as "Number of Public Transport Vehicles in San Pedro, Cazaapa and Caaguazu Road Segments"	34 per vehicle km	42 per vehicle km
			New: Number of rural community beneficiaries with improved access to all weather roads in the three departments	**0**	**34,100**

Table 2 summarizes the changes that were made to the intermediate outcome indicators at the same time as the Project restructuring.

Table 2 – Restructuring of the Intermediate Outcome Indicators (IOIs)[8]

Intermediate Objective	**Original IOI**	**Baseline**	**Target**	**Changed or Added IOI**	**Baseline**	**Target**
Developing the institutional capacities of MOPC for strategic planning and management of the road network	**Number of laws, decrees and regulations related to the Project enacted**	**0**	**100% of two identified laws (2)**	Unchanged	Unchanged	Unchanged
	Creation of a road strategic planning unit (mentioned in PAD Annex only)	**No unit**	**Unit created, operational equipped, and trained**	Unchanged	Unchanged	Unchanged

[8] IOIs and associated targets used in the evaluation of intermediate objectives are bolded

	Road strategy and five year plan implemented (mentioned in PAD Annex only)	No strategy or plan	Monitoring of Physical and Financial Execution of the Road Strategy	Becomes a PDO Indicator (see Table 1) and retitled as "Road Strategy and Five Year Plan implemented"	n/a	n/a
	Road Monitoring system implemented[9]	**No system**	**Reports of % compliance with strategy**	Unchanged (but moved to IO1 2)	Unchanged	**System established**
	Strategy of internal and external communication on road maintenance implemented	**No strategy**	Reports on % of monitored roads published	Unchanged	Unchanged	**Communications plan prepared, implemented and reported. MOPC performance report prepared, published**
Promotion of efficient planning and adequate resource allocation for the rehabilitation and maintenance of the road network	**Km of road network maintained under level of service**	**0**	**960 km through the financing of the Project (out of a total of 2016 km from all funding sources**)	Unchanged	Unchanged	**642 km through the Project (out of 1271 km from all funding sources)**
	Percentage of compliance with standards of length contracted	**0**	**No target**	Becomes a PDO Indicator (Table 1)	Unchanged	**90%**
	Programming and follow up on results based	**Non-existent**	**Report on execution of budget for results**	Unchanged	Unchanged	Unchanged

[9] This IOI is best related to the first intermediate outcome. At the time of the restructuring it was related to the second as the definition of the PDO in the PAD was used.

	management and budget implemented [10]					
				New indicator: **Master plan for optimization of roads toll collection developed and implemented**	**Outdated toll collection system with 16 stations**	**Modern integral toll collection system built and equipped for 11 stations**
Improvement of access of excluded rural communities to the primary paved road network in selected Departments	**Km of unpaved road network improved**	**0**	**338 km through the financing out of the Project (out of 876 km to be funded out of all sources in total)**	Unchanged	Unchanged	**140 km through the Project**
	Km of unpaved road network maintained at appropriate standards	0	893	Dropped	n/a	n/a
				New Indicator; **Indigenous Peoples plans prepared and implemented**	**No**	**IPP prepared and implemented with 3 multi-use centers**
				Maintenance system for unpaved roads is established through	**0**	**10**

[10] This IOI is actually best related to the second intermediate outcome. At the time of the restructuring it was related to the first as the definition of the PDO in the PAD was used.

				micro-enterprises in selected department		
				Refined indicator from an original PDO indicator and moved to IOI level: Vehicle operating cost – truck San Pedro, Caazapa and Caaguazu in PYG per km - veh	11483 (San Pedro), 11283 (Caazapa), 11360 (Caaguazu)	10682 (San Pedro), 10497 (Caazapa), 10569 (Caaguazu)
				Modified from an original PDO indicator: Number of public transport vehicles in San Pedro, Caazapa and Caaguazu (veh-km)	34	42
	Number of critical points of difficult transit	18	0	Dropped	n/a	n/a

1.4 Main Beneficiaries

The population of Paraguay as a whole was expected to benefit from the Project through its positive effects on competitiveness, employment and economic growth. Road users in general were expected to benefit from improved network quality and enhanced maintenance to reduce vehicle operating costs and travel times, especially on the road corridors to be improved under the Project. The direct beneficiary communities in the departments of San Pedro, Caaguazu and Caazapa were expected to see their access to opportunities improved through the secondary road development in their areas as well as through the expansion of job opportunities in the roads and other sectors. Finally through the development of Indigenous People's Plans (PPI) for each department, the specific cultural and economic needs of these groups were considered.

1.5 Original Components

The Project had three components, broadly corresponding to each of the three specific objectives (as stated in the PAD at least) as follows (see Annex 2 for full description):

(a) Strengthening Strategic Planning and Road Management (total cost US$6.5 million, of which US$4.55 million from the IBRD Loan);
(b) Improvement and Maintenance of the Paved Road Network (total cost of US$43.3 million, of which US$30.34 million from the IBRD Loan); and
(c) Improvement and Maintenance of the Unpaved Road Network (total cost of US$42.9 million, of which US$30.03 million from the IBRD Loan.[11]

In effect, the IBRD Loan was to finance all foreign costs (70%) and Paraguay was to cover the local costs inclusive of all taxes and duties (30%).

1.6 Revised Components

There were no revisions to the Project components.

1.7 Other significant changes

Five Project restructurings have been carried out principally but not uniquely, to progressively extend the closing date of the IBRD Loan as follows:

(a) June 2012 – for the purpose of extending the loan closing date to February 28, 2014 (to enable completion of Project activities[12])
(b) March 2013 – for the purpose of reallocating IBRD loan proceeds mainly for civil works (funds allocated from the "Unallocated" category and distributed according to requirements to complete Project activities[13])

[11] The components are titled somewhat differently in the Results Framework in the Annex to the PAD, without however there being any substantial difference in content of outputs and activities. The titles are respectively: strengthening MOPC's strategic planning and management; primary paved road network rehabilitated and maintained at an adequate level of service through performance based contracts; and unpaved national routes and secondary roads connecting rural communities to national network rehabilitated and maintained

[12] Project start-up had been slower than expected, but implementation had started to improve and was rated moderately satisfactory by June 2012.

[13] Revised allocation of funds was as follows (in USD million): Goods 2.540 (unchanged); Works 63.030 (was 52.400); Consulting Services 7.380 (was 9.580); Operating Costs 1.050 (was 0.400).

(c) January 2014 – for the purposes of: (i) extending the loan closing date to June 30, 2015[14] and (ii) revising PDO indicators to better fit the PDO.
(d) June 2015 – for the purpose of extending the loan closing date to December 31, 2015[15]
(e) December 2015 – for the purpose of extending the loan closing date to June 30, 2016[16].

The cumulative closing date extensions amount to 4 years, from the original June 30, 2012 to June 30, 2016.

2. Key Factors Affecting Implementation and Outcomes

2.1 Project Preparation, Design and Quality at Entry

Project Preparation and Readiness for Implementation

The Project was strongly linked to national objectives and CAS strategy. The PDO speaks to the contribution of the Project towards improving economic growth and alleviating poverty through improved road management strategy. The Project appeared to be well integrated into the framework of support provided by the Bank to Paraguay under the CAS. The Project's particular contributions to the country's objectives were through the promotion of sustained growth, especially in the rural areas, and to the enhancement of social inclusion, and through improving the coverage and efficiency of basic social services.

The Project was carefully defined over an extended preparatory phase. Three preparation missions were carried out in 2004, as were a further two in 2005. The Bank provided significant financial resources to the preparation phase, inter alia through the proceeds of a PHRD grant. Throughout the preparation process, the broad outlines of Project design were maintained. However one important element – that is the government and transparency improvement plan (IGAP) – emerged quite late in the preparation process. This factor was compensated by the additional attention given to this relatively new subject area for MOPC, including workshops carried out in March 2016 ahead of Project appraisal.

[14] Project implementation having suffered further delays partly due to administration changes and partly due to delays in start of main civil works and consulting services contracts.

[15] Additional delays were encountered in completion of civil works and finalization of social safeguards mitigation measures.

[16] The final extension was primarily for the purpose of permitting finalization under the IPP of the construction of community centers in the three departments, which was finally achieved at or just following the closing date.

Assessment of Design

The Project had a strong, upfront focus on establishing a sustainable road management strategy. Primacy was given to the development of a Road Strategy to be the principal underpinning of a more rational and well-resourced plan for the sector than had been the case in the past. This can be considered important in that the emphasis on the early development of the road sector strategy aligned with the priority accorded to it in the PDO. Similarly, many of the key institutional and supporting legal steps were identified during preparation and slated for early implementation

Project design was generally appropriate, but with some important provisos concerning the results framework. The three broad components of the Project-covering strengthening strategic planning and road management; improvement and maintenance of the paved road network; and improvement and maintenance of the unpaved road network - align well with the overarching objective of establishing a sustainable road management strategy. The components match up well with the three intermediate objectives under the PDO. The implied prioritization of these objectives is appropriate, starting from capacity improvement through adoption of key improvements affecting the main network through to complimentary attention being given to enhancing accessibility.

The original results framework does not match well with the PDO and the intermediate objectives (see Section 2.3). This did create difficulties to effectively track performance against PDO, especially in the earlier years of the Project. A number of the original KPI, such as reduction in vehicle operating costs, increase in public passenger transport, and reduction in transportation times, are not specifically relevant to the PDO and the critical expected outcomes of the Project. However in the case of the development of MOPC capacity the KPI and intermediate indicators, such as the creation to the road planning unit and the implementation of a road strategy, appear to be appropriate and measurable during the life of the Project. There was also a mismatch between the wording of the PDO in the LA and in the PAD resulting in the misallocation of two intermediate outcome indicators as mentioned in Table 2 above. In addition the target kilometers for the performance based maintenance contracts and the unpaved road improvements were not clearly delineated as those to be funded through the Project - reference is made to broader targets depending on actions and financing outside of the scope of the Project.

The Project's risks were generally identified and correctly rated. The overall risk rating for the Project was "High". The following specific elements were identified: the risk of varying Government support as ministers and key officials change; the risk created by working through a sector institution with weak technical and human capacity, and the risk associated with the lack of prior knowledge and experience in working with private sector performance based maintenance contracts. Also mentioned were the associated risks to Project implementation arising from limited knowledge in Bank procurement capacity and from the potential inadequacies or delays in local funding. As discussed below however, the strategies adopted by the Bank and the Borrower to mitigate these risks during implementation were not always effective.

The Project was likely to require significant enhancement to weak local capacity. The creation of the UPGP as part of the design can be considered an important step towards strengthening the core structures for project management. Beyond that however, the Project required significant institutional strengthening, not only in planning but also in more specialized areas such as performance contract (GMANs) management, micro-enterprises, tolls management and governance and transparency. Project preparation recognized the past difficulties encountered in assuring sufficient, stable financing for the sector, but perhaps paid insufficient attention to developing the measures required to resolve this. This inevitably had an adverse impact on local capacity to prepare and launch implementation of the Project.

2.2 Implementation

Overview

Implementation presented significant challenges throughout the life of the Project. The IBRD Loan was extended on several occasions in order to provide the space and opportunity required to complete Project activities. The original closing date was end June 2012, whereas the final completion date was end June 2016, largely due to later than expected start- up of the civil works activities. There appear to be many contributing factors to these delays, including recurrences in shortfalls in finance and personnel and also the impact of changes in government administration, as discussed below. The Project remained challenging to implement within the available resources and experiences of MOPC, given these included new forms of works contracts such as GMANs and micro-enterprises as well as an expanded social agenda, including governance and transparency. A number of contracts experienced delays in both in terms of procurement and implementation, due partly to limitations on knowledge and experience of Bank policies and procedures, especially in the cases of procurement and social assessment.

The Bank on balance underestimated these challenges, but nonetheless provided a lot of support towards improving implementation. Many of the challenges and the risk of delay associated with them could readily have been anticipated, as they were included under the appraisal risk assessment. The Project introduced a number of institutional changes associated with new initiatives in road planning and management. The capacity to absorb such changes was limited by knowledge and experience in the sector, exacerbated by the impact of changes in administration which occurred at end of 2008 and 2012 respectively.

The Bank however did field frequent supervision missions in the initial years of Project implementation and provided a list of prescribed activities for regular, mostly at least annual review, can be seen as providing a useful framework for getting Project

implementation on track after the slow start[17]. The Project however had very low disbursement due to delays in contracting activities through at least 2011, and it would be have been appropriate to downgrade the ratings at an earlier stage to draw attention to Bank management and the Government to the problems encountered.

The Bank was though effective in assisting in the development of an action plan in September 2011 which helped improve Project performance through a specific focus on action steps in four major areas: MOPC capacity improvement; acceleration of the GMANS contracts; acceleration of the unpaved road improvements; and increasing IBRD Loan disbursements. Implementation was generally more effective from this time on. This however was not reflected in the ratings for implementation progress which were for the most part "moderately unsatisfactory" up to Project closure. This rating appears to reflect largely the problems encountered in completing the required social mitigation measures, as discussed below. There were no problems of a comparable magnitude otherwise.

Factors generally within Government Control

The Project suffered a long delay in loan effectiveness. The IBRD Loan was finally declared effective in January 2008, whereas this had initially been targeted for January 2007. Three technical missions of the Bank were fielded between Board approval in July 2016 and the date of declaration of effectiveness. As there were no particular conditions of effectiveness, these delays can be attributed fully to the length of time taken to secure the necessary approvals and legal authority for the government of Paraguay to commit to the Loan.

Early stages of implementation were adversely affected by financial and personnel shortfalls. The start-up of Project activities was slower than anticipated in the implementation plan established at appraisal. Not only in 2007, but also in 2008 and 2009 by which time the Project had become effective, the Government did not provide all the budgetary resources required for the Project to be implemented. This can be seen as a critical factor in two senses: firstly because all Project activities were co-funded to the amount of 30% by counterpart resources; and secondly because these shortfalls contributed to delays in the appointment of key MOPC staff and also consultants critical to the implementation of various Project activities.

[17] The activities were as follows: follow the Operation Manual; carry out road and traffic survey (annual); carry out supervision of IPP and social participation framework; monitor IGAP (annual reviews and semi-annual reports on alert indicators); follow the procurement plan; implement and maintain a financial management system; provide financial reports (semi-annual); carry out project monitoring (semi-annual); and carry out an end of project evaluation (end of project).

Changes in government administration had some adverse effects on progress of the Project. The elections that took place towards the end of 2008 brought in a new administration and ended 60 years of rule by the Colorado party. Thus the transition that took place was more profound than any previous recent transition and more time was required to effect changes in ministerial and key official appointments, including those in MOPC. This transition happened at a time when the Project was just about to start implementation. The transition at the end of 2012 also had a significant impact in that further delays of 12-15 months occurred in launching the Project activities and associated institutional strengthening that had been identified at that time.

Factors generally within Implementing Agency control

Procurement activities suffered early delays. Even taking into consideration the delays in Project effectiveness, very little advance activity, especially in regard to procurement had taken place by end 2007. 2008 saw little progress made up to August, by which time election preparations had kicked in. Consequently, the implementing agency was not in a position to launch any bidding for works until the middle of 2009. These delays do appear to have been mostly tied to internal regulations – which factor appears to have been steadily mitigated over the course of the Project – rather than due to lack of knowledge or capacity. As a consequence, the GMANS contracts only got under way in September 2009 (GMANS 5), March 2010 (GMANS 1) and in December 2010 (GMANS 0 and 2)[18]. In the same vein, the contracts for the improvement of unpaved roads commenced only in May 2013 (San Pedro), June 2013 (Caazapa) and August 2013 (Caazuagu).

Changes in design contributed to higher than anticipated capital costs. The GMANs contracts were let based on initial basic designs, with the responsibility of the winning bidders to provide proposed final designs for MOPC approval. In this process, the amount of rehabilitation works tended to increase and thus estimated capital costs were higher than at appraisal. This situation seems to have arisen partly due to the fact that the initial designs dated from before the Project was approved and from the fact that the rate of deterioration of the pavement was greater than anticipated. A further contributory factor was that MOPC might be favorable to an approach that would lead to an improvement in the quality of the road over a maintenance-only intervention.

2.3 Monitoring and Evaluation (M&E) Design, Implementation and Utilization

Design

[18] The contract for GMANS 2 was subsequently canceled

Measurement of Project Results was not defined at appraisal Particular attention was paid to the higher level of objectives and to a methodology by which the impact of the Project on the broader goals of economic growth and poverty alleviation could eventually be assessed. The Project should however have provided for appropriate arrangements for carrying out the socio-economic impact analysis as this appears to be well beyond the experience and competence of MOPC. For many of the KPIs in the results framework, no baseline values were provided and insufficient attention appears to have been given to ensuring that this data was subsequently collected, say as a condition of effectiveness. In regard to both the socio-economic impact analysis – which could have provided very useful input for an assessment of road user satisfaction and beneficiary impact – and for the mentioned KPIs, final target data was not planned to be collected until after the conclusion of the Project. Therefore, this information was going to be of little value in measuring performance during the life of the Project.

Some of the original PDO indicators were not relevant - other PDO and intermediate indicators were missing . A number of issues occurred in the specification of the indicators, arising effectively from the incoherence of the results framework with the PDO. The original PDO indicators did not align very well with the PDO, with the exception of the one related to the creation of the road strategy. For example, the indicators relating to vehicle operating costs and, transportation times would have been relevant to a road improvement, but much less to one oriented toward a sustainable (maintenance based) road management strategy. None of the original PDO indicators appear to directly address promotion of efficient planning and resource allocation or improvement of access to excluded rural communities. In fairness however, there were a number of useful intermediate indicators, some of which were later to be provide the basis of revised PDO indicators under the January 2014 restructuring. Even after that, there were still some noticeable gaps, specifically no indicator at either level to track resource mobilization for road sector expenditures. Additionally, it would have been helpful to have an indicator to more explicitly track achievement of transparency and governance under the IGAP.

Implementation

The January 2014 restructuring addressed some, but not all, of these weaknesses. The restructuring correctly recognized and exploited an opportunity to review the results framework and generally came up with a number of improvements which were helpful in the sense of sharpening performance monitoring in the latter years of the Project. The resulting results framework aligned better with the PDO than did the original version[19] Under the restructuring, of the original PDO indicators only one was (partially) retained; four indicators were reassigned as intermediate indicators and one indicator was dropped altogether. The scope of one indicator - increased level of road user satisfaction was modified and its scope reduced in terms of geographical area. Four new indicators were

[19] Though it did not address the inconsistency between the LA and PAD versions of the PDO.

introduced, one of which had originally been an intermediate indicator. Following the restructuring and with the revised five PDO indicators, monitoring is assessed to have improved, as the indicators better reflected the expected outcomes and all had measurable targets and baselines.

The restructuring took place some 6 years after the declaration of effectiveness and was carried out ahead of the decision for a second extension of the closing date of the loan – from February 28, 2014 to June 30 2015. Monitoring of the Project up to that point in time had been less than effective as for the most part no information was being collected in regard to the PDOs. The situation improved after the restructuring, though this could be seen to be due not only to results framework improvements but to the imperative of completing Project activities prior to closing.

Use

The UPGP appears to have been generally diligent and effective in preparing regular reports on Project progress. These reports were generally useful in providing input for Bank supervision review and the preparation of action plans for improvement of Project performance. These reports were normally focused on implementation issues and, at least since January 2014, on compliance with PDO and intermediate indicators. The UPGP seems to have been less effective, as could also be said of Bank supervision, in generating and utilizing data in regard to the broader impact of the Project on beneficiaries. Sporadic attention was paid to trying to measure some critical indicators, such as vehicle operating costs and public passenger transport, with the result that end of the Project information on these indicators had not been collected by close of Project.

2.4 Safeguard and Fiduciary Compliance

Social Assessment. The involuntary resettlement (OP 4.12) and indigenous peoples' (OP 4.10) policies were triggered. Timely completion of all the required actions was an issue during project implementation, despite efforts made to ensure general compliance with the safeguards under all works contracts as well as to ensure effective communication with all affected parties. Generally speaking, capacity to manage the main social impact of road projects improved during the life of the Project. In specific regard to involuntary resettlement, there were very few issues arising. One exception was that there were some cases of illegal occupation of the right of way that had to be addressed. There was some land take involved with the new toll stations (specifically at Ita and Yparacai) and with the unpaved road component. With one of the unpaved road sections (San Pedro Lot 1), there was a specific issue concerning the routing of the improved road. This generated significant dissension among and between the served communities contributing to a delay in the completion of the works. The implementation of the Indigenous People's Plans (PPI) suffered significant delays in the implementation of some sub-components, specifically the construction of three indigenous training centers associated with each department (San Pedro, Caaguazu and Cazaapa), these only being completed by mid-2016. An issue for future sustainability will be that the funding for continued operation

and maintenance of these centers falls outside of the purview of the road sector. However a plan for the operational phase has been prepared and the roles of the *Instituto Nacional del Indigena* (INDI) and the municipal councils among others have been spelled out. It would likely have facilitated the PPI had the implementation arrangements provided for specific contributions from the national and departmental agencies responsible for indigenous people's issues at an earlier stage. This may have helped, for example, to avert delays associated with INDI's initial disagreement on the plans on where to locate and how to operate the centers.

Environmental Assessment. The environmental assessment (EA) safeguard (OP 4.01) was triggered. No significant or unanticipated environmental impacts arose during works' implementation. Generally speaking, capacity to manage the main environmental impacts of road projects was improved during the life of the Project. Delays in the issuance of environmental licenses for works under the Project were a factor during implementation – though there was some improvement in this regard over the course of time. A further concern were the delays that occurred in certain cases in the issuance of the final environmental reports (REFICAs) by MOPC, as this factored into the completion of handover of works by the contractors.

Financial Management. The Project did at times - early on at least - suffer from delays in the submission of reports and from shortcomings in the justification of eligible expenditures. The most recent audit reports have however contained clean opinions on the Project accounts, though drawing attention to the need to sharpen the analysis of non-eligible expenditures for the Project. Bank supervision of the financial management aspects of the Project was effective.

Procurement. The Project suffered some initial delays in contracting of works and consulting services due to lengthy and balky internal procedures. In regard to specific aspects of procurement, notably in regard to performance based contracts for which no standard documents existed at the outset of the Project, there were some capacity issues and lack of knowledge and experience of the relevant Bank policies and procedures. These shortcomings were mitigated to a significant extent by training and orientation provided by Bank specialized staff during implementation. Bank supervision of the procurement management aspects of the Project was effective. A concern towards the end of the Project was that there was no procurement specialist in place in MOPC, though this appears to have been subsequently and satisfactorily addressed.

2.5 Post-completion Operation/Next Phase

A follow up Transport Connectivity Project (Loan No. 86380-PY, P147278) in the amount of US$100 million was approved July 25, 2016. The operation builds on the lessons learned from the Project, particularly in terms of the design and implementation of performance based (rehabilitation and) maintenance contracts, and benefits from the capacity built in MOPC in this area. The selection of roads for the new project derives from the planning and strategic work carried out under the Project, especially under the planning unit (DPV) which came into being as a result of the Project. Sustainability and

benefits to be perceived by road users and the community at large will be enhanced by the wider geographical scope of the new project and by a new area of focus on road safety. The new project is expected to continue the process of deepening MOPC capacity for planning and management, especially in regard to maintenance.

The Country Partnership Framework (CPF) for 2015-2018, Report No. 82487-PY has been approved, with the specific goal of supporting the National Development Plan (NDP) in regard to poverty reduction, especially in rural areas. The CPF pillars are poverty reduction; inclusive economic growth; and international connectivity. The recent growth record in Paraguay has been impressive, but inequality remains a challenge. Improved road corridors remain vital to prospects for increased production and trade, through improved logistics and reduced transport costs.

3. Assessment of Outcome

The evaluation of Project outcome is based on the version of the PDO in the LA. The key objective statement to be evaluated is "to establish a sustainable road management strategy" whose result would be the upgrading and maintenance of the national road network as measured by three specific intermediate objectives as follows: strengthening MOPC institutional capacity, promotion of efficient planning and adequate resource allocation for rehabilitation and maintenance of the road network, and improvement of access of excluded rural communities to the primary road network in the mentioned departments. The intermediate objectives are considered to represent the three main dimensions of the sustainable road management strategy and thus are the key objectives evaluated to determine Project outcome.

3.1 Relevance of Objectives, Design and Implementation

The Relevance of the PDO is rated as High

The PDO was highly relevant to the development and sectoral challenges faced by Paraguay and to the strategies adopted by the government to address them. The PDO focused both on improvements to resource allocation, especially in regard to maintenance of the road network at both national and departmental level, as well as on strengthening sectoral strategy and institutional capacity to sustain road management. The expected impacts in terms of improved logistics, reduced transport costs and reduced social exclusion remain highly relevant to Paraguay's present situation. The economy remains small and open and heavily dependent on agricultural production and foreign trade. The Project maintained a balance between economic impacts and social impacts, with the latter including the contribution towards improved governance and transparency in a sector supported by major public expenditure. The PDO is also consonant with the recently approved National Development Plan (NDP) for the period 2014-2030, which seeks to address sustainable economic growth and poverty reduction over the medium term in Paraguay.

The Relevance of Project design is rated as Substantial

The PDO to establish a sustainable road management strategy links well to the three immediate objectives which are the means to bring about upgrading and maintenance of the road network. The Project's three components align well with the intermediate objectives in terms of their overarching intent. The structure of the Project allowed for a strong institutional strengthening component, much of which was successfully implemented early on in the Project, allied to highway rehabilitation and maintenance and improvement of access all of which are very relevant to the establishment of a sustainable road management strategy. Project outputs contribute directly to the achievement of the intermediate objectives. The allocation of funds under the Project to the three main components is broadly appropriate, though some reallocation was required in view of the higher costs incurred in carrying out the GMANs contracts which had a higher rehabilitation component than originally anticipated. The results framework had its shortcomings, especially at the level of the original PDO indicators which was addressed to a large extent later in the Project life. The intermediate indicators were broadly well designed (with perhaps a couple of omissions) and provided a basis for assessing intermediate outcomes.

3.2 Achievement of Project Development Objectives

The outcome indicators for the PDO were changed significantly following a January 2014 restructuring, as discussed in Section 1.3. Achievement is assessed against the targets for these revised indicators. Most of the original PDO indicators are not considered to be well aligned to the PDO as discussed in Section 2.1 and 2.3 above. In regard to the intermediate outcomes and associated targets, the evaluation also considers what was retained following the January 2014 restructuring. There were fewer changes than was the case for the PDO indicators, but in 3 cases an original target was changed and the resultant impact is discussed below in the relevant section.

PDO – Establishment of a sustainable road management strategy Rating: Substantial

The Project sought to support the establishment of a road management strategy which was to be a substantial improvement on what went before it in the sense of: an overarching focus on maintenance and preservation of the road network; the development and sustaining of knowledge and capacity in road sector planning; the introduction of methods, policies and procedures to increase efficiency of operations and resource use; and effectively integrating the various levels of the road network, with particular regard to the rural communities enjoying limited access to social and economic opportunity.

In overall terms, the Project is evaluated as having substantial achievements in regard to the PDO. The changes that have been supported through the Project – with its limited available resources - provide a good foundation for further future improvements which will enhance the quality of the road network and the effectiveness of all sources of financing for the sector. This is commendable the more so bearing in mind the weak

starting capacity of MOPC taken together with a political and administrative environment that had not been historically supportive of the type of changes contemplated through the Project.

As summarized below, all five indicators can be assessed as having been achieved. Notably, the road strategy and five-year plan (2008-2012) provided a platform for further plan updates (from 2013) and input to the National Development Plan (through 2032). Budget execution – for which an indicator was added in 2014 – has shown an improving trend. In the final year of the project was 89% of planned expenditure compared to a target of 75% and to an outcome of 83% in 2015. This is indicative of greater realism in estimating work output and better planning of the use of the resources made available. These achievements are buttressed by overarching improvements in national budget management that occurred during the life of the Project, such as clearer distinction between capital and recurrent spending.

There is one proviso to this assessment. In the case of the percentage of users reporting satisfaction with the quality of the roads, the actual values derived from a May 2016 survey comfortably exceed the baseline and target values. However there is a "halo effect" in play as the improvements in the road condition have just been realized as this has likely influences the high rating.

Table 3 – Summary of Achievement of PDO

Indicator	Baseline Value	Target Value	Actual Value
Number of rural community beneficiaries with improved access to all weather roads (in 3 departments)	0	34,100	34,100 according to the number of beneficiaries within 10 km of the improved roads
Increased level of budget execution	68%	75%	89% covering all programs carried out by MOPC, including but not limited to the Project
Percentage of users reporting satisfaction with the quality of the roads (in 3 departments)	14% good, 35% fair and 51% bad	50% good, 50% fair	77% good, 17% fair, 7% bad based on a May 2016 survey carried out on a sample of 637 beneficiaries
Road strategy and five year plan implemented	No strategy or plan in place	Strategy and plan in place	Road strategy was formally approved in December 2016. The five year plan was approved in 2007 and has been implemented
Percentage compliance with standards of the length contracted	0	90%	93% which is the weighted average on the three GMANS contracts carried out and financed under the Project.

Intermediate Objective I: Strengthening of the institutional capacity of MOPC.

The development of institutional capacities has helped to underpin the establishment of the road strategy. Planning, financing and management improvements have been to sustain strong sector performance. At appraisal, specific mention was made of the need to assure results focus, accountability and transparency. In this respect the Project can be rated as successful in that: performance based contracting has been introduced; contract management tools have been created; and improved public access to road sector information has been achieved.

Institutional improvements have been supported by legal actions have been taken in five areas: passage of transit and road safety law; passage of law to amend the classification of road system; creation of a planning department in MOPC; increase in toll rates to cover costs of the system; and creation of a department for transparency and public participation.

Through the Project an important advance has been made in the direction of creating and sustaining a modern road management structure and system in Paraguay. Though progress has at times been hampered by lack of staff and financial resources, MOPC has been able to develop some of the key building blocks for road planning – notably the road inventory and regular traffic counts as part of the road monitoring system – as well as developing the required institutional structures for mainstreaming of planning in key decision processes. Additionally in areas where the Project devoted resources to support innovation, notably in regard to performance based contracting and the use of micro-enterprises, management structures have been created to try to ensure continuity and sustainability.

Table 4: Intermediate Objective 1 Achievement Summary

Relevant PDO Indicator	**Intermediate Indicators**	**Target Outputs**	**Actual Outputs**
Road strategy and five year plan implemented (original, 2014 revision)[20]	Number of laws, decrees and regulations related to Project enacted	2 (Transit and Road Safety Law and Road Classification Law)	5 (including the two targets plus the decree to create the planning unit DPV, the toll revenue ministerial directive and the directive to create the transparency department DTPC

[20] In the original indicator the creation of the strategic planning unit was included and the five year plan was not.

	Creation of a road strategic planning unit	Unit created, operational, equipped and trained	Unit was created in February 2007 and has adequate funding and personnel
	Strategy of internal and external communication on road maintenance implemented	Communications plan prepared, implemented and reported. MOPC performance report prepared, published	Strategy fully under implementation since 2014. This includes but is not limited to the IGAP which is the critical tool for sector governance and transparency
	Road monitoring system	System established	System established, regular measurement of traffic and road condition in place.

Intermediate Objective II: Promotion of efficient planning and adequate resource allocation for the rehabilitation and maintenance of the road network.

A critical dimension of the Project concerns how available resources have been more efficiently used through planning in general and specifically to the increased use of performance based management. Results-based budget management has been put into place and regular reports are prepared to ensure MOPC can keep on track in relation to its budget execution goals. In regard to incremental resource mobilization for road maintenance, the main output has been the implementation of a master plan for toll modernization which is intended to improve revenue collection from road users on the tolled sections of the network. As the works were completed towards the end of the Project, there is no data yet available to quantify this impact.

The introduction of performance based contracts, known as GMANS, counts as an important outcome of the Project. Ultimately 3 contracts were implemented under the Project for a total of 624 km, whereas the original target had been 4 for a total of 930 km. All works under the 3 contracts (GMANS 0.1 and 5 respectively) were completed and the roads improved are under regular maintenance. A proviso is that in some cases more rehabilitation was required on the roads than had been expected, so that costs were generally higher than estimated and some contracts experienced delays to completion. This appears in part to reflect differences in the conditions on the ground compared to the designs.

More than 50% of all primary roads are now under some form of PBC, including all the road sections originally financed under the Project. This is in line with expectations at the Project outset that additional contracts would be funded under other sources, including but not limited to the 629 km under contracts financed by IDB in parallel to the Project. To this number can be added a further 600 km under new GMANS contracts

(317 km under IDB funding and 284 km under local funding) and at least 280 km of roads to be covered under GMANS2 (such as section of Ruta 3 and 4) which required a more significant rehabilitation intervention under external financing outside of the Project.

Table 5: Intermediate Objective II Final Achievement Summary

Relevant PDO Indicator	Intermediate Indicator	Target Output	Actual Output
Percentage of compliance with standards of the length contracted. (2014 revision)	Km of paved network maintained under level of service	624 km after 2014 restructuring	623 km completed under the three contracts GMANS 0, 1 and 5. GMANS 2 was rescinded.
Increased level of budget execution (2014 revision)	Programming and follow up on results based management and budget implemented	Results based programming and budgeting in place, with report on execution of project results	Results based programming and budgeting in place with report on execution of project results
	Master plan for optimization of roads toll collection developed and implemented	Modern integrated toll system built and equipped with 11 stations	System is in place, with a view to increasing road user revenues and maintaining high road quality

Intermediate Objective III: Improvement of access of excluded rural communities to the primary paved road network in selected Departments (San Pedro, Caaguazu and Caazapa).

In terms of the beneficiary population for the proposed improvements in the three departments, it is highly likely that the target number of 34,100 for improved access has been achieved. The expectation is that improved access: increases transport modal choice, reduces journey times and reduces operating costs. Improvements in mobility tend to lead to increases in access to employment opportunities and to social services.

The Project was able to deliver improvement of about 141 km of unpaved roads, together with a number of bridges, in the three departments. This was in line with the target revised as of January 2014 - the original was for 338 km covering proposed interventions within the Project based on the original budget and funds allocation. This target had to be significantly reduced as the cost of the GMANS contracts was much higher than anticipated. Delays were experienced in the contracting and execution of the works. In the case of Caaguazu, drainage requirements were underestimated and designs had to be changes as a result. In the case of Caazapa, works were seriously impacted by issues connected to poor soil quantity and proneness to flooding.

Due to these issues, contracted costs were generally higher than estimates and additional rehabilitation works had to be incorporated. There is a limited record of achievement to date in regard to the maintainability of the roads given the completion delays. The

Project anticipated that the improvements in the rural roads would both stimulate an increase in the provision of public transportation as well as a reduction in vehicle operating costs. These impacts are expected to be positive, but may likely be experienced some time in the future. To date there is limited information in this regard.

The Project was able to support the introduction of micro-enterprises which were intended to be an efficient and socially relevant solution to routine maintenance, though the target number was not achieved. This nonetheless represents an important innovation for Paraguay, likely to be sustained given the capacity improvements in MOPC, giving rise to increases in employment and incomes in beneficiary communities.

The strong social dimension for this objective is underpinning by having the completion of social assessments including indigenous peoples plans drawn up for each of the three departments. The required actions did experience various delays during implementation, but these have helped to create the conditions for effective, future civic participation at departmental level.

Table 6: Intermediate Objective III Final Achievement Summary

Relevant PDO Indicator	**Intermediate Indicator**	**Target Output**	**Actual Output**
Percentage of road users reporting satisfaction with the quality of roads in San Pedro, Caaguazu and Cazaapa (2014 revision)	Indigenous Peoples Plan prepared and implemented	IPP prepared an implemented with 3 multi-use centers	Plans completed for each of the 3 departments, including the construction of the 3 community centers
	Maintenance system for unpaved roads established through micro-enterprises	System established with 10 MEs	System established with 6 MEs and ready for mainstreaming as resources permit
Number of rural community beneficiaries with improved access to all weather roads in the three departments (2014 revision)	Km of unpaved road network improved	140 km following the 2014 restructuring	143 km of road improvement completed in the 3 departments

Note: The intermediate indictors for vehicle operating cost reduction and increase in public passenger vehicles are not included here. In neither case is there data available for the post Project situation.

3.3 Efficiency

Rating: Modest

Overall, returns to the use of funding provided through the Project are positive, though lower than estimated at appraisal. The aggregate ex ante estimate economic rate of return was 33% whereas the estimate ex post ERR is 21%

The main contributory factors have been: higher than estimated capital costs as many of the road sections covered required more rehabilitation than had been expected to sustain the targeted level of improvement in roughness; and delays in both the procurement and the implementation of works which resulted in the postponement of benefits to be derived from the roads improvements. However these factors have been partly offset by stronger than forecast traffic growth, estimated at 7% per annum compared to 2.5% ex ante, which have helped to drive higher quantified benefits.

For the GMANS contracts, economic analysis was carried out using the HDM-IV model which simulates life cycle conditions and costs. For the ex ante analysis, vehicle fleet characteristics were defined and economic costs were estimated at 85% of financial costs. The analysis was carried out over a 15 year period with an adopted discount rate of 12%. Traffic was derived from data recorded on toll stations over the period 1995-2004. The ex ante analysis considered three sections that were originally set for the financing by the Project, for a total of 968 km.

For the ex-post analysis, the same methodology was used but considering the actual costs of the works carried out as well as the estimated actual traffic growth. Only data for GMANS 0 and GMANS 1 were calculated given that GMANS 2 works were not in the end carried out under the Project. The actual capital costs for the rehabilitation and periodic maintenance elements of the contract were identified, as were the actual construction periods. The total length of roads improved was 489 km. Traffic was estimated using 2011-2014 data for the same toll stations used in the ex ante analysis. In the absence of new VOC numbers, the original estimates were used, but updated for inflation.

Further to the table below, the weighted average ERR for the two sections post evaluated is 42% compared to the aggregate ex ante estimate of 41%.

Table 7: Economic Evaluation of GMANS Contracts[21]

	GMANS 0		**GMANS 1**		**GMANS 2**	
	NPV (US$mn)	ERR (%)	NPV (US$mn)	ERR (%)	NPV (US$mn)	ERR (%)
Ex Ante	14.7	60	8.7	31	7.3	32
	22.4	74	11.4	28	n/a	n/a

[21] No ex ante economic analysis was prepared for GMANS 5

For the unpaved road improvements, the ex- ante analysis used the RED model which is adapted from HDM IV to conditions prevailing on low traffic roads. The analysis covered a total of 338 km of road sections in the departments as follows – Caaguazu 122 km, San Pedro 113 km and Cazaapa 103 km as well as the bridges to be improved in each department. This was considered to be a "first year work program" with a capital cost of $12.6 mn.

Under the Project however a reduced works program was implemented than envisaged at appraisal with a total length improved was 140 km but at a capital cost of $24.6 mn, indicative of a change in the scope of works compared to what had been envisaged. For the ex post analysis consideration was given only to those links which retained for the program. No separate ex post analysis has been carried out for the bridges whose costs have been included in the revised estimated capital costs of the road sections improved.

Further to the table below, the calculated ex post ERR is 15% compared to 32% for the ex ante analysis.

Table 8: Economic Evaluation of Unpaved Roads

	Caaguazu	San Pedro	Caazapa	Total
Unpaved Roads				
NPV(US$mn)				
Ex Ante	3.12	6.38	4.70	14.20
Ex Post	(0.83)	8.79	1.14	9.10
ERR(%)				
Ex Ante	24	39	36	32
Ex Post	11	20	13	15

The economic analysis of the toll optimization program (cost of US$3.3 mn) considered the benefits of the incremental toll revenue collection realized over a period of five years. Investments were expected to expand toll operations and increase efficiency. The "with project" and "without project" income and cost streams were compared with a discount rate of 12%.

It has not been possible to compile an ex post analysis, as the implementation of the toll modernization program suffered delays (only fully completed in April 2016) and there are no data on the actual "with project" income streams. There is however a risk that the returns to the program will be less than originally estimated as the final capital cost of the works was significantly higher than estimated, $11.9 mn compared to $3.2 mn. However, as with the other elements of the works program, this will likely be partially offset with the higher than projected traffic volumes which should lead to higher revenue streams.

3.4 Justification of Overall Outcome Rating

Rating: Moderately Satisfactory

To arrive at a final overall outcome rating, relevance, efficacy and efficiency have been evaluated separately as above and the scores have been combined according to the Bank ICR guidelines. The derivation of the final outcome as *Moderately Satisfactory* is set out in the table below.

Table 9: Summary Evaluation of Project Outcomes

	Rating
Relevance **PDO** **Design**	High Substantial
Efficacy **PDO**	Substantial
Efficiency	Modest
Overall Rating	MS

3.5 Overarching Themes, Other Outcomes and Impacts

(a) Poverty Impacts, Gender Aspects, and Social Development

The Project sought to address social exclusion in rural areas through adopting a participatory process for the selection of roads for improvement in the three selected departments. This has been supported by the implementation of specific measures under the PPI which have helped to increase inclusion and enhance responsiveness to previously marginalized communities.

Civic participation in the planning and management of the road sector in Paraguay has been more generally assisted by the implementation of the IGAP. The program has created new monitoring tools and radically increased the public availability of information on the road network. The design of the web site has enhanced accessibility through the use of graphic interfaces. There is also provision for complaints and other feedback to be provided to MOPC.

Action in regard to gender was an integral element of the program to promote micro-enterprises for routine maintenance on the unpaved network. Female participation in these micro-enterprises – now at 26% of total - was enhanced through encouraging

participation and stressing the overarching social development goals to which income generated from employment on routine maintenance could contribute.

(b) Institutional Change/Strengthening

Within the framework of a public sector ministry responsible for public works, the Project was instrumental in promoting significant change and strengthening, based on the adoption of a new road management strategy. This assistance was oriented first and foremost to enhancing the basic planning and management functions, but also covered innovation in contracting, strengthening of toll management and revenue generation as well as the integration of a number of critical activities linked to enhanced governance and transparency in the road sector. These changes were supported by the Government during Project implementation although the achievement of some of the original targets was delayed or incomplete, in particular regard to the road inventory.

The IGAP has proven to be a good instrument for introducing and deepening the institutional commitment to greater transparency. The Paraguay experience is seen as a model which has been shared very widely with other road sector institutions in the region. Particular mention can be made of the very effective tool for monitoring contracts that was developed. Starting from a quite basic model, the functionality of the tool and the general accessibility of information was steadily improved over time.

Overall however it should be recognized that the Project played a leading and critical role in the transformation of the maintenance culture in Paraguay through the innovation introduced through performance based contracting. By the end of the Project more than 50% of the primary road network was under some form or other of multi-year works contract, managed on the basis of achieving and maintaining a level of service to the road user. Also all road sections that had been covered by GMANS contracts under the Project continued to receive attention for maintenance - and rehabilitation where required - after the closure of the Project.

(c) Other Unintended Outcomes and Impacts (positive or negative)

n/a

3.6 Summary of Findings of Beneficiary Survey and/or Stakeholder Workshops

n/a

4. Assessment of Risk to Development Outcome

Rating: Moderate

Concerning the infrastructure improvements secured through the Project, it will be critical to sustain robust maintenance arrangements and the required funding to go with it. The introduction through the Project of performance-based contracting has helped to establish a stronger maintenance culture and organization which now appears to apply generally to the network regardless of funding source. This is very evident for the

national network, perhaps less so for the departmental network, though a good start has been made there with the introduction of the micro-enterprises. At the macro level, establishing and sustaining the right levels of funding may continue to be a challenge in the absence of dedicated funding flows other than tolls. However improvements in planning and management will certainly assist in using the resources available as efficiently as possible.

Concerning institutional development, the Project made good inroads into a substantial, ambitious agenda that had been set out at the outset. The effectiveness of MOPC has clearly been enhanced with institutional capacity built in a number of areas. The approval of the IBRD funded follow on Transport Connectivity Project will help assure the provision of further support for institutional strengthening, with particular regard to road management where some activities, such as road inventory, remain to be completed. The implementation of the IGAP has helped in regard to the development of new instruments to better monitor contracting and works performance as well as to provide a conduit for better public information and civic participation. The accessibility of information and the quality of its presentation has improved over the life of the Project, providing a firm foundation for future initiatives in this area.

In regard to social development, a continued, strong relationship with INDI and the local authorities will be needed. The Project was successful in developing a comprehensive social management framework, including an IPP. Implementation of the IPP was subject to significant delays, due in part to difficulties encountered in identifying and carrying out mitigation measures in the three departments. There was clearly an element of learning here for the road sector and MOPC in particular. But going forward, the eventual achievements cannot be seen as an end in itself but a stepping stone to a wider, more concerted effort to manage the social aspects of road maintenance. In this regard relationships with INDI and the local authorities will have to be sustained and improved.

5. Assessment of Bank and Borrower Performance

5.1 Bank Performance

(a) Bank Performance in Ensuring Quality at Entry

Rating: Moderately Satisfactory

The Bank provided significant assistance to the Government in putting together a balanced, relevant operation that in terms of design responded well to the national economic and sector issues and priorities of the time.

The two areas of concern that mitigate this generally favorable assessment for the Bank which are: firstly, the institutional change and development agenda was a very challenging one that might better have been broken into phases to avoid delays in implementation; and secondly the design of the results framework had gaps and flaws, especially the lack of measurable targets for most PDO indicators, This latter did create

difficulties to effectively track performance against PDO, at least in the earlier years of the Project.

(b) Quality of Supervision
Rating: Satisfactory

The Bank fielded an extensive number of supervision missions in the early stages of the Project, though these were not always effective either in addressing delays in effectiveness or in accelerating implementation and disbursements in the first three years. It might have been prudent to downgrade the Project earlier than was the case and to bring on board restructuring at an earlier stage in the life of the Project.

During the later years of implementation,the supervision effort found more success, especially after the development of the September 2011 action plan. The efforts made can be judged as successful insofar that most activities under each of the Project components appear to have been carried out and largely completed. Following a restructuring of the results framework in January 2014, most targets for the PDO and intermediate outcome indicators could be met. But this did require to extend the Project closing date on several, separate occasions to accommodate earlier delays to implement agreed actions..

(c) Justification of Rating for Overall Bank Performance
Rating: Moderately Satisfactory

The overall rating of *Moderately Satisfactorily* reflects that, whilst overall due care and attention were afforded to the Project in all stages, some opportunities to enhance Project impact were missed at both design and implementation phases.

5.2 Borrower Performance
(a) Government Performance
Rating: Moderately Satisfactory

The Government provided support at the Project preparation stage and later was a partner with the Bank in trying to address the causes of delays in implementation ahead of Project closing. There were delays in achieving Project effectiveness and the early stage of implementation were adversely effected by a number of changes in key ministerial and senior appointments. There was also recurring theme of lateness and at times insufficiency of counterpart funding for the Project. This appears to have been more effectively mitigated at later stages in Project implementation.

(b) Implementing Agency or Agencies Performance
Rating: Moderately Satisfactory

There were major initial delays in Project implementation related to slowness and complexity in the bidding process for contracts financed under the Project, for which the

agency has to take some responsibility. In fairness though, the agency had to navigate a steep learning curve given the number of obligations they were faced with, a number of which required structural change and building capacity in areas new to MOPC. Given these circumstances, the agency acquitted itself well and was able eventually to deliver, albeit with some delay, major contract works including performance based contracts which were introduced to Paraguay for the first time under the Project.

(c) Justification of Rating for Overall Borrower Performance
Rating: Moderately Satisfactory

The overall rating is *Moderately Satisfactory* is derived from the balance of the assessments for government and implementing agency. This reflects that despite difficulties encountered in securing the continuous, efficient implementation of the Project, over the course of the Project there was some success in addressing the capacity and financial constraints. The Government did maintain its support to ensuring that the Project could be completed in all its aspects.

6. Lessons Learned

a. **Performance based contracting can provide the critical underpinning for the strategy to improve and sustain road maintenance.** A pivotal feature of the Project was the introduction of PBC, using long term maintenance contracts for the first time in Paraguay. This can be judged as successful in terms of making a solid start on the development of a sustainable maintenance culture, something that had not been true of Paraguay historically. The experience suggests that performance contracting can be successfully introduced into low capacity environments with the proper planning and addressing of constraints. MOPC has now developed the capacity and experience to plan and manage this type of program and is building towards the use of PBCs for more complex, longer term rehabilitation and maintenance contracts (CREMA).

b. **Sectoral governance and transparency programs have the potential to play an important role in strengthening of road planning and management**. The initiatives taken through the IGAP were well founded in terms of addressing public perception and civic participation. But they also directly and positively impact road management, enhancing the efficiency and effectiveness of investments through improvements to monitoring and problem resolution. Through the Paraguay experience, a very effective too for monitoring contracts was developed. Starting from a quite basic model, the functionality of the tool and the general accessibility of information was steadily improved over time. These programs should and could be replicable in other countries with however duel attention needing to be paid to: ensuring the timetable is not overoptimistic and allowing needed time to build capacity as well as environment for improvement; and maintaining a healthy balance between completing works and improvements in infrastructure with transparency and openness to public participation.

c. **Attention has to be paid at Project design to ensure the risk assessment is truly aligned with the implementation schedule.** Both the Bank and the implementing agency are subject to pressures from their respective stakeholder communities to deliver projects with complex objectives within relatively short time frames. However it would be important to recall that implementing a challenging institutional change agenda will tend to run into obstacles along the way. It would be advisable to assess with due care and attention all dimensions of the change process to identify and try to mitigate specific risk factors. There will also be the need to give reasonable accommodation to the impact of political and administrative change along the way. This Project provides examples of the risks to design and implementation of complex and/or new project initiatives in a low capacity environment. These risks include the learning process on environment and social mitigation measures which can be much longer than estimated upfront.

d. **In project design, specific care and attention needs to be given to defining clear, measurable and achievable indicators.** It is necessary, but not sufficient, that a project adequately and clearly address national and sectoral goals. For effective project monitoring however, it is critical that the results framework include clear, measurable indicators with baseline and target values. In this instance there were deficiencies in the design and coverage of these indicators, which could and should have been addressed much earlier than they were eventually, that is at the time of the January 2014 restructuring. It is important that the indicators fully reflect all of the most important benefits of the project, as otherwise there is a risk that the evaluation of results may miss certain key factors. .

e. **Where maintenance funding is such a critical consideration for sustainability, the project needs to ensure attention is properly paid.** The project in this case rightly looked at measures for improving efficiency and effectiveness of sector institutions to improve maintenance performance as well as for the aforementioned introduction of PBCs. However it was clear that these efforts could only go so far when there was persistent underfunding of maintenance needs by government, a situation only marginally mitigated by direct funding of users. This suggests that more attention has to be given to this factor in project monitoring and that consideration ought to be given in project design to the inclusion of indicators and targets to track these critical financing flows.

f. **Financing of contracts through joint contributions of the Government and the Bank should be avoided.** One of a number of factors in play in regard to delays in project implementation in this case was the financing in the ratio of 70:30 from the IBRD Loan and from Government. This exacerbated delays in contract payments when the two sources of funding could not be aligned. There is no overarching reason to maintain this situation given that all the funds are ultimately the Government's. It would have been better to finance contracts either 100% by the IBRD loan or 100% by other sources. In certain contemporaneous Bank funded projects, the initial co-financing arrangements have been modified through a project restructuring, which would have been an avenue to explore in this case.

7. Comments on Issues Raised by Borrower/Implementing Agencies/Partners

(a) Borrower/implementing agencies

n/a

(b) Cofinanciers

n/a

(c) Other partners and stakeholders
(e.g. NGOs/private sector/civil society)

n/a

Annex 1. Project Costs and Financing

(a) Project Cost by Component (in USD Million equivalent)

Components	Appraisal Estimate (USD millions)	Actual/Latest Estimate (USD millions)	Percentage of Appraisal
Strengthening Road Management and Strategic Planning	6.51	7.42	114
Improvement and Maintenance of the Paved Road Network	43.35	73.47	169
Improvement and Maintenance of the Unpaved Road Network	42.90	26.34	61
Total Baseline Cost	92.75	107.23	116
Physical Contingencies	8.79	0.00	0.00
Price Contingencies	3.69	0.00	0.00
Total Project Costs	105.23	107.23	102
Front-end fee IBRD	0.19	0.00	0.00
Total Financing Required	105.42	107.23	102

(b) Financing

Source of Funds	Type of Cofinancing	Appraisal Estimate (USD millions)	Actual/Latest Estimate (USD millions)	Percentage of Appraisal
Borrower		31.42	38.51	122.00
International Bank for Reconstruction and Development (IBRD)		74.00	72.00	97.00
Total		105.42	110.51	105.00

Annex 2. Outputs by Component

1. Strengthening Strategic Planning and Road Management (projected total cost of US$6.51 million, estimated actual total cost of US$7.42 million)

The planned activities under each of the four sub-components - taken together with the principal outputs and associated intermediate indicators under the Project - are set out below:

Sub-component 1.1: ***Strengthening of Strategic Planning;*** This subcomponent was to finance consulting services, goods and training for the following:

(i) improve the legal framework of the road sector and road classification;
(ii) set-up a strategic planning unit and equip it with the necessary instruments; knowledge base and training for it to begin performing its tasks;
(iii) introduce and implement a traffic counting system;
(iv) strengthen the procurement unit with a software system to improve its contract and procurement monitoring; and
(v) provide strengthening training for the VMOP.

Key outputs are as follows:

Laws and regulations passed – 5 as opposed to target of 2 – as follows: Transit and Road Safety Law passed in 2014; Road Classification Law passed in 2016; creation of the DPV in MOPC by decree in February 2007; toll revenue ministerial directive (source of funding for road maintenance); and ministerial directive to create the transparency department, DTPC, in February 2007. **Intermediate Indicator**

Create strategic planning unit – the enabling measure was taken in February 2007 prior to loan effectiveness. DPV created, operational, equipped and trained **Intermediate Indicator**

Under IDB funding, the road sector investment plan was prepared and the transport master plan updated based on inputs provided through a consultancy by Consorcio NK-NKLAC.

A road planning and management system has been in place since September 2015, updating of the five year investment plan is in progress. A major consultancy input was provided by Consultora Bicentenario for US$0.45 million.

Traffic counting system is in place since October 2014 as part of the road monitoring system. A major consultancy input was provided by CSI-Ciemsa in the amount of US$1.85 million. **Intermediate Indicator**[22]

[22] This indicator though appears in the revised results framework under Component 2.

Sub-component 1.2: ***Strengthening Budget Preparation and Execution;*** This subcomponent was to finance consulting services to:

(i) develop a management of works and budget execution monitoring system; and
(ii) develop and implement an automated works certification and payment system.

Key outputs are as follows:

Budgeting and programming based on results is in place. Pilot program for results based management budget is being implemented. A report is issued twice a year to the Ministry of Finance. **Intermediate Indicator**

Project financial management tools are in place

Sub-component 1.3: Strengthening Road Management and Maintenance: This subcomponent was to finance consulting services, training and goods to:

(i) strengthen management of road improvement and maintenance contracts;
(ii) update the pavement management system and the road inventory; and
(iii) develop and implement a communication strategy on road maintenance;

Key outputs are as follows:

A communications strategy has been in place since 2014. A communications plan has been prepared and is under implementation. An annual progress report from the Minister is published. A monitoring tool for project execution has been developed and implemented. **Intermediate Indicator**

A road inventory was carried out and mostly completed during the Project. Key consultancy input was provided by RF Consultoria y Asociados in the amount of US$0.42 million.

Sub-component 1.4: Monitoring and Evaluation of the Project: This sub-component was to finance works and consulting services to set-up base lines and mechanisms to monitor outcome and impact indicators. These activities were to include:

(i) measuring vehicle operating costs;
(ii) socioeconomic impact surveys, baseline and monitoring;
(iii) baseline and surveys of user satisfaction;
(iv) strengthening and training UGPG in project coordination and Bank procedures etc.:
(v) implementation of Indigenous People's Plan; and
(vi) implementation of Environmental Mitigation Plan;

Key outputs include the carrying out of training, completion of user satisfaction and baseline surveys and the completion of the indigenous people's plan and environmental mitigation plan.

2. Improvement and Maintenance of the Paved Road Network (projected total cost of US$43.35 million, estimated total actual cost of US$73.47 million)

Sub-component 2.1: Performance-based Maintenance Contracting (projected US$39.67million, estimated actual cost US$60.37million): This sub-component was to finance:

(i) a training program on performance-based contract preparation and monitoring for MOPC staff; and
(ii) contracts that included initial rehabilitation works to meet the minimum level of service, road safety works, vertical and horizontal signaling, and maintenance works of the national paved road network, through performance-based maintenance contracts (GMANS). The prospective road networks were identified and contracts were structured into seven GMANS each of five years duration covering much of the Eastern region of Paraguay.

The project was to finance four of the identified GMANS for a total length of 968 km as follows: GMANS 1 (381km): covering routes PY02, PY03 and PY08; GMANS 2 (344 km): covering Southern Access to Asuncion, route PY 01, PY 04, and GMANS Urban[23] (108 km): covering part of route PY02 and part of Southern Access to Asuncion, and GMANS 5 covering the San Ignacio – Pilar section of PY04 (135 km).

The intention was that MOPC would seek financing from other sources to cover additional contracts on 1,086 km of paved roads, but following the procedures and mechanisms developed by the project.

In the event, the IBRD Loan financed only 3 out of 4 of the planned contracts that is GMANS contracts 0, 1 and 5 for a total of 623 km. Works were finalized and signed off before end 2015. GMANS 2 was adjudicated, but canceled as the contractor failed to comply. The adjusted performance target for roads improved was substantially met.

GMANS 0 (108.2 km) covered two sections as follows (1) Ruta Acceso Sur Cuatro Mojoenes to Acceso Guaranbare (35.2 km) and (2) Ruta PY02 San Lorenzo to Caacupe (73 km). The contract was awarded to Talavera and Ortellado with a final cost of US$5.47 million for an implementation of 1440 days. There were no cost and only a minor time overrun (works were stopped at some point due to a payment issue). Supervision was provided by Consorcio CIA - Grimaux in the amount of US$0.72 million.

GMANS 1 (381 km) covered three sections as follows: (1) Ruta PY02 from the Arroyo Irala bridge to the start of the Tape Para concession (127.6 km); (2) Ruta PY03 between Limpio and the conection to PY08 (147.7 km); and (3) Ruta PY08 between the connections to PY02 and PY03 (105.7 km). The contract was awarded to Consorcio de

[23] Later known as GMANS 0

Ingenieria Vial in a final amount of US$17.97 million, with an implementation period of 1440 days. There were no cost or time overruns. Supervision was provided by Consorcio Engevix in the amount of US$0.88 million.

GMANS 5 (135.5 km, but initially 134 km) covered Ruta PY04 San Ignacio to Pilar. The contract was awarded to Consorcio Vial Sureno in a final amount of US$32.73 million with an implementation period of 1,085 days. There was a cost overrun of 29% and a time overrun of 73%. Supervision was by Consupar-Esteio for US$1.71 million, an overrun of 95%

In the case of GMANS2, a mobilization payment (only) was made to contractor Diferencial Tape in the amount of US$0.54 million. Supervision was by Consorcio Electropar-Applus in the amount of US$0.29 million.

Sub-component 2.2: Toll System Optimization (projected US$3.6 million, estimated actual US$13.1 million): This sub-component was to finance:

(i) a Toll System Optimization Master Plan, and

(ii) investments to implement the Master Plan. The Toll System Master Plan would ultimately define which tolls would be relocated, what sequencing would be used to start charging in both directions and what investments in infrastructure and systems are necessary. The loan would finance part of the Master Plan, concentrating on the tolls that are on the routes that will benefit from a performance-based contract.

The following contracts were executed under the Project:

Civil works carried out by M&T SA – Traffic LTDA in a total amount of US$11.87 million.
Supervision provided by D y B – CCT in the amount of US$0.94 million.
Consultancy carried out for the elaboration of the toll master plan by Consorcio CSI – Geipix in the amount of US$0.21 million.

The related **intermediate indicators** for this component are as follows:

a. Paved road kilometers under PBCs – 623 km meets revised target
b. Tolls master plan in place and implemented – complete by early 2016, with official inauguration in June 2016. 11 new toll stations have been instructed with significant improvements in operationality and efficiency compared to the existing system of 16 stations (which is being phased out)
c. Road monitoring system in place – the system is contracted and under implementation. 1,955 km of paved roads and 5,000 km of unpaved roads have been surveyed. 15,786 km non paved roads have been inventoried. The inventory was not completed at the end of the life of the Project however.

3. Improvement and Maintenance of the Unpaved Road Network (projected total cost of US$42.90 million, estimated actual total cost of US$26.43 million)

Sub-component 3.1: Road improvement and maintenance works. The road sections to be improved under this sub-component were determined following an initial ranking of all the Departments in the country on the basis of objective criteria (such as rural population, poverty, number of kilometers of unpaved network). Following this analysis, the top three ranked Departments were selected: Caaguazu, San Pedro and Caazapa. The candidate road network in each Department was then chosen through a transparent participatory process to determine the priority segments in each Department.

In the event, contracts were implemented for the improvement of about 141 km of unpaved roads as summarized below.

Roads in San Pedro were awarded in in 3 lots as follows:

A (Chore-Colonia Cocuera-San Pablo - 35.5 km) awarded to Vialtec for US$ 6.12 million (a 29% overrun) for an implementation period of 720 days (a 20% overrun).
B. (Chore – Jhugua Rey - 29 km) awarded to Benito Roggio y Hijos for US$ 4.50 million and 600 days with no cost or time overruns.
C. (Jhugua Rey – Gral. Aquino - 25.5 km) awarded to Ecomipa for US$ 4.39 million and 600 days with no cost or time overruns.

Supervision was by Consorcio Caem-Geolatina for US$0.74 million.

Roads in Caaguazu (Colonias Unidas – Jhoy Central – J. Raul Oviedo – J. Eulogio Estigarribia) were in one contract for 35.7 km awarded to Consorcio Roggio-Technoedil for US$5.88 million (a 14% overrun) for 564 days (a 25% overrun).

Supervision was by Consorcio Consultec - Mercoconsul for US$0.31 million

Roads in Caazapa roads (Caazapa to Boqueron) were in one contract for 14.7 km awarded to Consorcio Numi for US$3.73 million (an overrun of 18.5%) for 686 days (a 90% overrun).

Supervision was by CIA-Grimaux for US$0.23 million.

Under a contract with MYO, multipurpose community centers were constructed in each of the three departments. The centers were created following the agreed action plan arising from the Indigenous People's Plans (PPI) prepared by Red Comunitaria Guarani.

In regard to maintenance, a total of 6 micro-enterprises have been created covering total of 93 km. One micro-enterprise is all indigenous. The average female participation rate is 26%. The program was designed with the assistance of the ILO with which organization a contract of US$0.336 million was signed.

Sub-component 3.2: Capacity Building and Development Program for MOPC District Offices and Regional and Local Governments. This subcomponent was to finance works, goods and consultant services to
(i) improve the infrastructure, systems and equipment in MOPC District offices to improve road management and contacting of maintenance;
(ii) build capacity for MOPC District staff on managing maintenance contracts; and
(iii) build capacity for planning and management on roads maintenance for Regional and Local Governments.

Key outputs included the provision of improved infrastructure and equipment and trained personnel at the level of district offices.

5 **intermediate indicators** are related to these outputs as follows:

a. Non paved roads improved – 141 km
b. IPPs developed and implemented – yes, 3 different ones
c. Micro-enterprises in place – 10 was the target but only 6 created to date
d. VOC for trucks reduced - (still awaiting final measurement), target is (in PYG per veh-km, 10682, 10496 and 10568 for respective departments.
e. Increase in public transport from 34 to 42

Annex 3. Economic and Financial Analysis

At appraisal, the MOPC performed the economic analysis of the civil works project components: (i) performance-based maintenance contracts; (ii) improvement of unpaved roads; and (iii) improvement of bridges. The measured project benefits were savings in vehicle operating costs and travel time costs and maintenance costs as a result of improved ride quality and travel times. For paved roads, net benefits were evaluated using the Highway Development and Management Model (HDM-4) Version 1.3, which simulates life cycle conditions and costs and provides economic decision criteria for multiple road design and maintenance alternatives. For unpaved roads, net benefits were evaluated using the Roads Economic Decision Model (RED) Version 3.2, which adopts the same economic analysis principles and road user costs as HDM-4, being tailored for the economic evaluation of unpaved roads.

Th ex ante overall economic rate of return (ERR) was found to be 33%. The aggregate ex ante estimates are given below:

Table 10 Aggregate Ex Ante Economic Returns

	Benefits (US$mn)	Costs (US$mn)	NPV (US$mn)	ERR (%)
Performance-based Contracts	37.9	7.3	30.6	41
Unpaved Roads	23.7	8.8	14.2	32
Bridges	8.9	6.6	3.0	22
Total	70.5	22.7	47.9	33

For the ex-ante economic analysis, vehicle fleet characteristics and economic costs were defined for seven vehicle types, economic costs being 85% of financial costs. The analysis period was set to 15 years and the adopted discount rate was 12%. The expected annual traffic growth rate was defined based on the observed annual traffic growth from 1994 to 2004 on seven toll stations and was set to 2.5% per year for all vehicle types for the duration of the analysis period. Road work costs were estimated in financial and economic terms, economic costs being 89% of financial costs.

The ex-post economic analysis of the project was done for the performance-based contracts and for the improvement of the unpaved roads program, adopting the same methodology as the one applied at the ex-ante economic analysis and the same assumption, but considering: (i) the actual costs per km of the road works; (ii) the better quality of the road works executed on the unpaved roads; and (iii) an estimate of the actual annual traffic growth over the project period.

Because the roads rehabilitated or improved under the project were not necessarily the same as the ones evaluated at appraisal and on some roads the road works length changed, the ex-post analysis considered the actual average costs per km of the road works of each contract, comparing them with the appraisal estimates.

Performance based contracts

Of the three performance-based contracts evaluated at appraisal, two of them were financed under the project (GMANS1 and GMANS 0), which were reevaluated for the ex-post analysis. The table below presents the comparison of the capital investment costs per km, which are the costs of the periodic maintenance or rehabilitation works under each contract. For GMANS 0 the actual cost per km were 73% higher than the appraisal estimates, while for the GMANS 1, the actual costs per km were 15% lower than the appraisal estimates.

Table 11. Performance Based Contracts Capital Investment Costs

		Total	Investments	Capital Investment	
		Length (km)	Length (km)	Total (M US$)	Total per Km (US$/km)
PAD	GMANS 0	108	86	2.22	26,018
Estimate	GMANS 1	381	120	4.20	35,026
ICR	GMANS 0	108	108	4.87	45,013
Actual	GMANS 1	381	381	11.30	29,658
Actual/	GMANS 0				1.73
Estimate	GMANS 1				0.85

Unpaved Roads Improvement

The improvement of unpaved roads program improved roads in three departments. The length of the road works financed under the project reduced from the appraisal estimates and the unit costs increased in part to the due to the change in the quality of the executed improvement works. The table below presents (i) the road works length evaluated at appraisal and the executed road works length; (ii) the appraisal cost estimates per department and the actual costs per contract[24] in each department; and (iii) the resulting road improvement costs per km, excluding the bridge improvement costs. On all three departments the road improvement costs per km increased compared to the appraisal estimates, being around five times the appraisal estimates. The ICR analysis estimated that the change in the quality of the executed improvement works reduced unit road user costs by around twice the reduction estimated at appraisal[25].

[24] Three contracts were financed on San Pedro, of which Lot 1 corresponds to the roads evaluated at appraisal.

[25] Corresponding to a reduction from 16 to 8 IRI compared to a reduction from 16 to 12 IRI.

Table 12. Improvement of Unpaved Roads Program Road Works

		Roads Length (km)	Roads Cost (M US$)	Bridges Cost (M US$)	Roads Cost per Km (US$/km)
PAD Estimate	Caaguazú	121.6	4.90	1.85	40,307
	San Pedro	113.4	4.20	1.19	37,016
	Caazapá	102.9	3.50	3.61	34,000
ICR Actual	Caaguazú	35.7	6.87	0.75	192,508
	San Pedro (Lot 1)	35.5	6.19	0.49	174,345
	Caazapá	14.7	3.26	1.27	221,576
Actual/ Estimate	Caaguazú				4.78
	San Pedro				4.71
	Caazapá				6.52

Ex Post Economic Analysis

The actual annual traffic growth over the evaluation was estimated based on the observed annual traffic growth data from 2011 to 2014 collected on sixteen toll stations and the GDP increase observed over the project evaluation period. The traffic on the toll stations increased at 8.1% per year from 2011 to 2014 and the GDP at constant prices increased at 5.7% per year over the same period, resulting on an elasticity of 1.4. Considering that the GDP at constant prices increased at 4.9% per year from 2007 to 2015, the ICR analysis estimated that the actual traffic growth rate from 2007 to 2015 was 7.0% percent per year, which is almost three times the annual traffic growth rate estimated at appraisal (2.5% per year).

The table below presents the results of the ex-post economic analysis, considering the actual traffic growth, actual costs per km, and actual reduction in unit road user costs for the unpaved roads. The overall ERR of the performance-based contracts remained about the same as the one estimated at appraisal, at 42%, considering that comparing to the appraisal estimates the ERR of the GMANS 1 decreased to 28% (higher actual costs per km), while for the GMANS Urban increased to74% (lower actual costs per km). The overall ERR of the unpaved roads improvement program reduced to a still satisfactory 15%, compared to the 32% estimated at appraisal, which is due to the higher actual costs per km, which were not compensated by the actual increase in traffic and actual reduction in unit road user costs. The overall ex-post ERR of the two project components evaluated for the ICR is 21%.

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Table 13: Ex-post Economic Evaluation Results

Program		Ex-Ante NPV (M US$)	Ex-Ante EIRR (%)	Ex-Post NPV (M US$)	Ex-Post EIRR (%)
Performance-based Contracts	GMANS 1	8.70	31%	11.41	28%
	GMANS 0	14.70	60%	22.41	74%
	Subtotal	23.40	43%	33.82	42%
Unpaved Roads Improvements	Caaguazú	3.12	24%	-0.83	11%
	San Pedro	6.38	39%	8.79	20%
	Caazapá	4.70	36%	1.14	13%
	Subtotal	14.20	32%	9.10	15%
Total		37.60	37%	42.92	21%

Toll Optimization Program

The ex ante economic analysis of the toll optimization program (cost of US$3.3 mn) considered the benefits of the incremental toll revenue collection realized over a period of five years. Investments were expected to expand toll operations and increase efficiency. The "with project" and "without project" income and cost streams were compared with an NPV of US$10.3 mn at a discount rate of 12% estimated as a result.

It has not been possible to compile an ex post analysis as implementation of the toll modernization program suffered delays and there are no data on "with project" income streams. There is a risk that the returns to the program will be less than originally estimated as the final capital cost of the works was significantly higher than estimated. However, as with the other elements of the works program, this will likely be partially offset with the higher than projected traffic volumes which should lead to higher revenue streams.

Annex 4. Bank Lending and Implementation Support/Supervision Processes

(a) Task Team members

Names	Title	Unit	Responsibility/ Specialty
Lending			
Andres Pizarro	Senior Transport Specialist		Team Leader
Jose Maria Alonso Biarge	Highway Engineer, Consultant		
Graciela Sanchez Martinez	Social Development Specialist		
Mary Lucy Giraldo	Procurement Specialist		
Rodrigo Archondo-Callao	Highway Engineer		
Antonio Blasco	Financial Management Specialist		
Alejandro Alcala Gerez	Counsel		
Veronica Raffo	Consultant		
Roberto Munoz	Environmental Specialist, Consultant		
Daniel Gross	Resettlement Specialist		
Lucio Caceres	Institutional Development Consultant		
Rodrigo Garcia Verdu	Economist		
Xiomara Morel	Senior Finance Officer		
Supervision/ICR			
Andres Pizarro	Lead Transport Specialist		Team Leader (to end 2010)
Maria Margarita Nunez	Senior Highway Engineer		Team Leader (2011-April 2014)
Stephen Muzira	Transport Engineer		Team Leader (April 2014 to April 2015)
Marcela Silva	Lead Transport Specialist		Team Leader (April 2015 to October 2015)
Maria Claudia Pachon	Senior Transport Specialist		Team Leader (November 2015 to October 2016)
Kavita Sethi	Senior Transport Economist		Team Leader (from October 2016)
Veronica Raffo	Senior Infrastructure Specialist		
Aracelly Woodall	Senior Program Assistant		
Virginia Henriquez Fernandez	Infrastructure Consultant		
Claudio Daniele	Environmental Specialist		
Graciela Sanchez Martinez	Social Development Specialist		
Graciela Von Bargen	Procurement Specialist		
Jorge Daniel Arguindegui	Procurement Specialist		
Gustavo Adrian Canu	Procurement Specialist		
Justina Duarte	Financial Management Specialist		
Natalia Bavio	Financial Management Specialist		
Daniel Chupowicz	Financial Management Specialist		
Juan Carlos Serrano Macharro	Financial Management Specialist		

Maria Pia Cravero	Counsel		
Ruth Gonzalez	Communications Specialist		
Stephen Brushett	Lead Transport Specialist		ICR Author

(b) Staff Time and Cost

Stage of Project Cycle	Staff Time and Cost (Bank Budget Only)	
	No. of staff weeks	USD Thousands (including travel and consultant costs)
Lending		
FY05	20.48	117,628.93[26]
FY06	36.70	200,035.66
FY07	14.56	78,304.32
FY08	6.38	19,428.21
Total:	**78.12**	**415,397.12**
Supervision/ICR		
FY07	3.01	12,446.33
FY08	9.19	35,231.91
FY09	37.17	155,487.30
FY10	19.29	169,902.43
FY11	31.77	237,003.48
FY12	22.99	159,228.53
FY13	21.98	200,479.68
FY14	21.93	146,683.74
FY15	18.26	119,172.56
FY16	17.04	111,309.24
FY17	4.30	22,132.88
Total:	**206.93**	**1,369,078.08**

[26] An additional amount of $930,240 was provided through PHRD trust funds

Annex 5. Beneficiary Survey Results
n/a

Annex 6. Stakeholder Workshop Report and Results

n/a

Annex 7. Summary of Borrower's ICR and/or Comments on Draft ICR

INTRODUCCIÓN

Este documento es el informe de cierre del Proyecto de Rehabilitación y Mejoramiento de la Red Vial del Paraguay "Ñamoporá Ñanderapé", proyecto que fue financiado con recursos de crédito de la banca multilateral, Banco Mundial-BIRF-PA-7406, y recursos del presupuesto nacional, ejecutado en el marco del Convenio suscrito entre el Banco Mundial y la República de Paraguay.

Los insumos fundamentales de este documento han sido los documentos que reposan en los archivos digitales y físicos del Proyecto dentro de las instalaciones del MOPC, información relevante que sirvió para compilar, resumir y analizar desde la concepción y puesta en marcha hasta el cierre y posterior evaluación del Proyecto.

El informe recoge el cierre de los componentes desarrollados durante la vida del Proyecto 7406 PA, ejecutados por el Ministerio de Obras Públicas y Comunicaciones, a través del Viceministerio de Obras Públicas y las Direcciones intervinientes como la Dirección de Vialidad y la Dirección de Planificación Vial, la Unidad Ejecutora GMANS creada a través del Préstamo y la Unidad de Preparación y Gestión de Proyectos (UPGP) unidad de gestión, supervisión y administradora del Préstamo 7406 PA.

Así mismo, se hace una recopilación y actualización de la evaluación de la ejecución del Proyecto, teniendo en cuenta el cumplimiento de los objetivos planteados, cumplimiento de los compromisos asumidos, cumplimiento de los compromisos del Banco Mundial, las lecciones aprendidas y las recomendaciones realizadas, la estrategia de sostenibilidad del Proyecto, el resumen de los archivos físicos y digitales del Proyecto, el cierre administrativo y financiero del Proyecto parcial a setiembre 2.016, y la actualización de las fichas de seguimiento y ejecución de proyectos.

ANTECEDENTES

El Gobierno de la República del Paraguay y el Banco Mundial han firmado el Convenio de Préstamo 7406-PA, para la elaboración del Programa de Mejoramiento, Gestión y Mantenimiento de la Red Vial del Paraguay. La administración y coordinación del Programa es responsabilidad del Ministerio de Obras Públicas y Comunicaciones (MOPC) del Paraguay, a través de la Unidad de Preparación y Gestión de Proyecto (UPGP).

El objetivo general del Programa es el de establecer una Estrategia de Gestión para la Red Vial nacional que apunte al uso racional de los recursos públicos destinados al mejoramiento y mantenimiento de las carreteras.

Los objetivos específicos del Programa son: a) Capacitar y brindar las herramientas necesarias al MOPC a fin de que mejoren la planificación de la red vial;

b) Prevenir el deterioro y mejorar el nivel de servicio a los usuarios de caminos que forman parte de las carreteras de integración internacional y regional que componen la red nacional pavimentada, (Rutas 1, 2, 3, 4, 7 y 8);

c) Mejorar la accesibilidad de las comunidades rurales más pobres a la red nacional pavimentada en tres departamentos del país para una efectiva integración económica y social, a través del mejoramiento y mantenimiento de caminos nacionales y departamentales no pavimentados, beneficiando a zonas con mayores necesidades de ocupación y empleo.

El Programa tiene por fin contribuir al crecimiento económico y la reducción de la pobreza del país, mejorando las condiciones de la infraestructura vial, el acceso rural, y el impacto del presupuesto público; promoviendo la transparencia a través del uso de mecanismos basados en el desempeño, disponibles para el conocimiento de todos los actores involucrados, y mecanismos participativos.

El Proyecto se conforma de tres componentes:

- **Fortalecimiento Institucional**, con el objetivo de desarrollar las capacidades y herramientas institucionales del MOPC para una planificación estratégica y gestión vial integrada y programática de la red vial;

- **Red Vial Nacional Pavimentada**, con el objetivo de prevenir el deterioro y mejorar el nivel de servicios a los usuarios de los corredores viales de integración internacional y regional que componen la red vial fundamental del Paraguay;

- **Caminos Nacionales y Departamentales no Pavimentados**, con el objetivo de mejorar el acceso desde zonas rurales (y particularmente desde las comunidades rurales más pobres y marginales) a la Red Vial Nacional pavimentada, para fomentar una efectiva integración económica y social del país.

El *objetivo principal* *del Proyecto* de establecer una Estrategia de Gestión para la Red Vial nacional se conseguirá apuntando al uso racional de los recursos públicos destinados al mejoramiento y mantenimiento de las carreteras y buscando (i) *transparencia*, mediante la planificación y programación de todas las intervenciones basadas en criterios objetivos y cuantificables; (ii) mejorar *el desempeño*; mediante la identificación de metas para cada intervención, que serán monitoreadas y cuyo cumplimiento será la indicación del éxito de la misma, (iii) *la divulgación*; mediante la participación, publicación, y presentación de las acciones programadas a los actores y a los beneficiarios involucrados, así como la publicación y presentación de los resultados obtenidos.

PARA LA SOSTENIBILIDAD DETALLE DE RESULTADOS POR COMPONENTE Y ASPECTOS CRITICOS

Objetivo 1: Fortalecer la capacidad institucional en el MOPC en la gestión y planificación vial

A través de este componente se apoyaron diversas iniciativas encaminadas a fortalecer el área de gestión y planificación vial. En particular vale la pena resaltar que a partir de la elaboración del Plan Maestro y del Plan Vial, se implementaron las obras identificadas como prioritarias para el periodo hasta el 2015. Como resultado de ello se cumplió con las metas previstas en el Plan: (i) incrementando del número. de Km Pavimentados de 4,000

km a 5,600 Km; (ii) incrementando del número. de Km de caminos mejorados (empedrados, enripiados) de 1,500 Km a 5029 Km; y (iii) se redujeron el número de Km de tierra.
Por otra parte, el proyecto también apoyo diversas iniciativas para el fortalecimiento del marco normativo del sector incluyendo entre las más relevantes:

- Ley de Tránsito y Seguridad Vial,
- Ley de categorización y clasificación vial
- Creación de la Dirección de Planificación Vial (DPV) en el MOPC
- Resolución Ministerial de Peajes que habilita al pago bidireccional así como el incremento del cobro de peajes como fuente de financiamiento para el mantenimiento de la red vial
- Resolución Ministerial que crea la Dirección de Transparencia y Participación Ciudadana (DTPC).

Contratos Adicionales Financiados

Adicionalmente, se financiaron los siguientes contratos que son fundamentales para el fortalecimiento de la planificación estratégica del MOPC:

- **Sistema de Conteo. Contrato concluido**

  Con personal capacitado y en etapa de operación. Como resultado de esta actividad, hoy en día están en operación 9 puestos permanentes (tráfico y carga), 14 estacionales (tráfico) y 4 puestos ocasionales. El relevamiento de datos y el procesamiento ya se hace permanentemente y se utiliza para efectos de planificación y diseños en diferentes áreas del MOPC.

- **Sistema de Gestión y Planificación. Contrato concluido.**

Objetivo.

Contar con un sistema de información que sirva de herramienta de decisión para una gestión de planificación de la infraestructura vial administrada por el MOPC, basada en información técnica económica y socio ambiental actualizado y confiable.
Con este Sistema se está haciendo la actualización del Plan Vial de los próximos 5 años.

- **Inventario Vial: Contrato inconcluso, al término del periodo de Préstamo se desarrolló 43 % del Inventario de la red vial del país.**

Objetivos

General.

Contar con la información básica para la gestión vial consistente en un inventario físico y de estado de los elementos que componen los caminos del Paraguay administrados por el MOPC.

Específicos.

Definir y precisar las características físicas y su estado de todos los caminos que forman la red vial administrada por el MOPC. Así podrá conocerla y valorarla para su adecuada administración con el objeto de preservar o mejorar esa infraestructura. Por ello se requiere contar con un inventario vial actualizado, completo y geo referenciado.

- **Desarrollo de une herramienta automática de monitoreo de la Ejecución Financiera del Proyecto.**

Se apoyó el diseño e implementación de esta herramienta la cual se encuentra disponible en el sitio web: http://www.mopc.gov.py/mopcweb/BM/IGAP/index.html desde junio de 2015. Esta herramienta sirve para el monitoreo interno y el público en general del progreso financiero del proyecto, detallando todos los contratos financiados por el proyecto en tiempo real.

Esta herramienta es el resultado de la implementación del Plan de Acción de Mejora de Gobernabilidad (IGAP) del Proyecto que fue diseñado para contribuir con la eficiencia, transparencia y mitigación de prácticas corruptas, incluyendo originalmente aspectos administrativos, financieros, adquisiciones, sociales y ambientales. Pese a que el IGAP había recibido elogios como un modelo de buenas prácticas para la mejora del ambiente de gobernabilidad en el sector de transporte, la atención a su implementación ha variado a través del tiempo y diferentes resultados fueron alcanzados. Paso de ser una herramienta en Excel para monitoreo interno y manual a uno informatizado alimentado por los datos de los procesos del MOPC en tiempo real.

La fortaleza de la herramienta radica en la independencia de entradas manuales de datos por cuanto se alimenta e integra datos del sistema informático de Presupuesto ya existente del MOPC y los datos abiertos de la Dirección Nacional de Contrataciones Públicas (DNCP) con las siguientes funcionalidades:

1. Gráfico de círculos que representan los contratos y adendas en donde el tamaño corresponde a su monto.
2. Categorización de contratos por: *i*) tipos de obras y servicios por componentes del proyecto, *ii*) rubros, *iii*) tipo de licitación y *iv*) proveedor
3. Evolución del estado de las contrataciones y ejecuciones en la línea del tiempo.
4. Búsqueda de contratos por palabras claves que aplica al gráfico de círculos y a la lista.
5. Listado de contratos que se pueden filtrar por código de contratación, nombre del llamado, RUC y razón social del proveedor adjudicado, monto del contrato y tipo de licitación.

6. Detalle de pagos por cada contrato con la fecha de obligación del pago, el concepto y el monto.
7. Descarga de datos en formato JSON para su utilización en análisis adicionales por desarrolladores de aplicaciones.
8. Asimismo, la herramienta cuenta con un sistema de generación de informes automáticos y con un enlace a la Ventana de Acceso a la Información(VEA) http://www.mopc.gov.py/mopcweb/BM/monitoreo.html

Oportunidades para replicar la herramienta. La herramienta desarrollada es potencialmente replicable a los demás proyectos del MOPC por cuanto que el código fuente del sistema podría adaptarse a las necesidades de cada proyecto. La replicación de la herramienta plantea costos razonables para los proyectos e incluyen:

- La adaptación de los procesos de extracción de datos de pagos de la base de datos del MOPC y los agrupadores de contratos se puede hacer con el equipo del MOPC.
- No se requiere de compra de nuevo hardware para su implementación, ya que el MOPC cuenta con servidores adaptables.
- No se requiere de licencias de software adicionales, ya que toda la tecnología utilizada para el desarrollo de la herramienta es de licencia abierta (*open source*).

Los detalles técnicos de la herramienta se encuentran en el Manual de Operaciones IGAP Web.

Objetivo 2: Evitar el deterioro de la red vial primaria y asegurar el mantenimiento de su nivel de servicio

Bajo este componente se apoyó el cambio en el paradigma de gestión de la red vial pavimentada, mediante la introducción de contratos por estándares y niveles de servicio. En este sentido se ejecutaron con recursos del préstamo tres contratos de obras de Gestión y Mantenimiento (GMANs) sobre una longitud total de 624,7 Km (GMANs 0,1,y 5).
En los tres contratos se cumplieron con los estándares de servicio definidos en el contrato.
El MOPC ha financiado con recursos de otras fuentes de financiamiento, contratos por niveles de servicio sobre una longitud aproximada de 1268 Km, con lo cual la cobertura del nuevo sistema alcanza aproximadamente 23% del total de la red pavimentada.
Con base en esta experiencia, la nueva estrategia de gestión de la red vial pavimentada del MOPC está llevando a cabo en el año 2016 la gestión de aproximadamente 50% (2,600 Km) de la red pavimentada bajo la modalidad de contrato modelo CREMA donde se contemplan intervenciones de recuperación mayores a las que se incluían en el sistema GMANS.
Esta política del MOPC es una clara señal no solo del impacto positivo que ha tenido el proyecto en mejorar la eficiencia de la gestión vial, sino también en la sostenibilidad de la inversión al largo plazo.

Objetivo 3: Mejorar el acceso de las poblaciones rurales más pobres en los Departamentos de San Pedro, Caaguazú y Caazapá.
Bajo este componente se financiaron contratos de obra para el mejoramiento de la red de tierra sobre una longitud total de 140,39 Km para mejorar la transitabilidad permanente sobre estas rutas.

Caminos No Pavimentados de San Pedro Fueron intervenidos tres tramos: (i) Camino B08 Choré – Colonia Cocuerá – San Pablo; (ii) Camino B09 Choré – Jhugua Rey; (iii) Jhugua Rey - Gral. Aquino, con un total de longitud para el Departamento de San Pedro de 90,00 km. Obra terminada.

Caminos No Pavimentados Caaguazú Fue intervenido un tramo: Colonias Unidas - Raúl Oviedo -Yjhovy Central - J. E. Estigarribia de 35,7 km. Obra terminada.

Caminos No Pavimentados Caazapá. Fue intervenido un tramo: Caazapá – Boquerón de 14,3 km. No ha concluido a la fecha.

Por otra parte, bajo este componente también se apoyó un programa de microempresas de mantenimiento de la red no pavimentada. El objetivo de este proyecto piloto era Sentar las bases conceptuales, normativas, institucionales y operacionales para la implantación de un Sistema de Mantenimiento Rutinario de Caminos con Microempresas en el Paraguay, con enfoque inclusivo, a cargo del Ministerio de Obras Públicas y Comunicaciones.

Se financiaron 6 microempresas cubriendo una longitud total de 93 Km. Bajo este programa se generaron 46 empleos directos en total de pobladores locales, de los cuales 29% son mujeres y una de las microempresas está compuesta por poblaciones indígenas. Cada microempresa cubre una longitud de entre 12 y 25 Km con 5 a 10 socios (2.5 km/socio). El costo promedio de mantenimiento rutinario bajo este programa asciende a US$2,500/km-año. El MOPC envío al Banco el informe de cierre de este piloto. Si bien el MOPC observo las bondades del programa, no se le ha dado continuidad al mismo.

DESEMPEÑO DEL BANCO
Calificación

S (Satisfactorio)

Según la perspectiva de la UPGP el desempeño del Banco es calificado por el prestatario como Satisfactorio.

Como consecuencia del largo período de implementación, siete gerentes de Proyecto han supervisado la ejecución del Programa.

Los funcionarios designados por el Banco contaban con la experiencia y conocimientos heterogéneos y en la mayoría de los casos han contribuido sustancialmente a solucionar aspectos críticos de la gestión del Programa y también a facilitar su evaluación.

Sin embargo en otros casos, la Unidad de Preparación y Gestión de Proyectos ha encontrado diferencias y, por lo tanto, el acompañamiento y apoyo que brindaron al Programa, se vio resentido por retrasos en el envío de las No Objeciones necesarias.

Vale resaltar que estos inconvenientes fueron, superados merced a los buenos oficios, capacidad y empeño de los funcionarios de ambas partes

En un balance global, ha sido destacable la asociación de trabajo que se generó entre los equipos técnicos de la UPGP y los funcionarios del Banco, durante todo el período de ejecución.

Uno de los principales puntos críticos que ha presentado el accionar del Banco, se sitúa en el atraso inicial y las excesivas demoras en la aprobación de las distintas contrataciones y/o licitaciones de Obras que retrasaron la ejecución de los proyectos.

En el último periodo del Préstamo la comunicación ha sido fluida por los distintos medios que se han utilizado como ser email, video conferencias y audio conferencias, lo que ayudó muchísimo a lograr los compromisos asumidos.

DESEMPEÑO DEL PRESTATARIO
Calificación:

S (Satisfactorio)

El desempeño de la Unidad de Preparación y Gestión, amerita un análisis por Etapa del Proyecto.
En un principio: La conformación de la UPGP se da netamente con funcionarios del MOPC, los cuales tuvieron que trabajar para la Aprobación del Préstamo en el Congreso y dado que eran de la función pública encontraron mucha disponibilidad con sus pares del Congreso, aunque de igual manera se tuvo un periodo de aproximadamente 2 años para la aprobación, en contrapartida por ser la primera experiencia de formación de una UPGP BM, no contaban con la experiencia en las Normas de Adquisiciones del BM, lo que dificultó la preparación de los documentos de licitaciones del Préstamo, tampoco se contaba con el personal para cada área.

Este acontecer se ve reflejado en la ejecución del Préstamo en los dos primeros años de implementación.

En una Etapa que denominaremos intermedia; se procede a la contratación de consultores de planta a fin de ayudar en la implementación y mejorar la ejecución, los consultores contratados fueron técnicos de las diferentes aéreas a ejecutarse, de esta forma se logra un fortalecimiento institucional de la UPGP y se forma un grupo de trabajo conscientes del compromiso contraído y con la ayuda que significó las capacitaciones realizadas por el Banco Mundial, se fueron dando más celeridad a las gestiones de los documentos.

En cuanto a los procedimientos internos al MOPC, las gestiones para las licitaciones tuvieron mucho retraso, debido principalmente a que los funcionarios del mismo conocen los procesos la Ley 2051, que regula las contrataciones nacionales; no así las Normas de adquisiciones de los organismos financieros internacionales como el BIRF.

En la Etapa final del Préstamo las capacidades adquiridas del personal consultor y funcionariado y la comunicación constante con el Banco Mundial para los procedimientos se fueron agilizando llegando inclusive a que las Licitaciones fueran publicadas con un ahorro del 50 % de tiempo invertido que al comenzar el Préstamo.

Actualmente se observan como aspectos positivos:

I. Pro actividad y mejoras en gestión y control de la ejecución;
II. aumentos de la eficacia en la ejecución de los proyectos.
III. Seguimiento y evaluación, en constante proceso de mejora
IV. Gestión financiera adecuada
V. Significativos progresos en las relaciones con otras instituciones nacionales y gobernaciones

LECCIONES APRENDIDAS Y RECOMENDACIONES

1. Respecto a los Contratos GMANS:

- El promedio de longitud de los GMASN interurbanos diseñados (1 a 6) ha sido de 317 km / contrato. Lo aconsejable es tener mallas más acotadas, desde ya dependiendo de las condiciones geográficas y características de las mismas, en el entorno de los 150 km de longitud promedio de la Malla. El GMANS urbano tuvo una longitud de 108 km, y resulta lógico una longitud menor en centros urbanos.

Dependiendo de las características geográficas el dimensionamiento podría estar en el entorno entre los 100 km y 200 km aproximadamente, con un promedio de 150 km. La inversión inicial de obras de puesta a punto también incide en la decisión de la longitud de las mallas de caminos que conformen una unidad económica rentable para el sector privado.

- Los tramos que conforman la malla del contrato GMANS deben estar agrupados por tramos homogéneos bien definidos, agrupados por rangos de tránsito, estructura del pavimento existente y condición de estado.

Para ello es necesario contar con un Informe de Investigación de la zona de obras que realice el MOPC para definir el denominado Proyecto Básico o Referencial de las OPP mínimas: i) relevamiento de estado, ii) medición de IRI, iii) deflexiones, iv) espesores, v) calicatas y clasificación de suelos, vi) estudios de tránsitos disponibles, vii) inventario físico para que el oferente evalúe lo faltante según normativa vigente (ejemplo señalización vertical, estado de alcantarillas, etc.).

- Para el umbral de IRI a exigir al final del contrato, deben hacerse una proyección de IRI correlacionado con modelos de deterioro como AASHTO, o bien utilizar el HDM-4 verificando condiciones de calibración.

- Se requiere que el PBC defina expresamente cuales son las Obras de Puesta a Punto, estas obras debe indicarse que son mínimas y que el contratista deberá evaluar a su riesgo las obras necesarias para poder cumplir con los umbrales establecidos, bajo el concepto de suma alzada y no pensar en el concepto de unidad de medida. Los pliegos establecen obras obligatorias y se debería adaptar al concepto de obra mínima necesaria para cumplir el estándar. Esta definición de las OPP debe incluir el riesgo de la carga circulante.

- Es importante definir un Precio Base acorde con un Proyecto Base bien estudiado con mediciones de la red, ya que es una disposición a pagar el Estado y una referencia para la definición posterior del monto del contrato. Para ello se requiere de un archivo estadístico, datos de inventario y necesidad de fortalecer el desarrollo de las capacidades institucionales del MOPC.

- Los estándares a exigirse en el DDL deben estar particularizados para cada uno de los tramos homogéneos que componen la malla de rutas del proyecto GMANS. Para ello debe definirse estándares de exigencias específicos para cada tramo homogéneo. También se recomienda que los tramos homogéneos sean de una longitud aceptable para el manejo del contrato (10 a 20 km, dependiendo de cada malla y por supuesto de las características del tramo).

- Definir un plazo de al menos 60 días hasta 90 días, para la presentación del Diseño Ejecutivo por parte de la Contratista, que incluya todas las mediciones técnicas e informes de ingeniería para cumplir con los estándares exigidos, y que definan las obras de puesta a punto, como resultado de las evaluaciones estructurales y no solo superficiales. Debe preverse un plazo en función de la longitud y complejidad del tramo.

- Ante el planteo de los Contratistas de la posibilidad de reubicar por tramos las cantidades mínimas según la necesidad, esto debe salvarse y definirse a la aprobación del Diseño Ejecutivo, según las justificaciones técnicas que presente el Contratista, con las mediciones e informes de ingeniería correspondientes.

- Resulta necesario la participación de la Fiscalización en el Diseño Ejecutivo, por desfasaje en el inicio de sus Servicios respecto a la Obra. En este sentido la Fiscalizadora debería estar contratada durante el proceso de adjudicación de la obra o bien, como se propone en el punto de Fiscalizadoras, contratarle proyecto básico y fiscalización.

- Se recomienda definir los plazos en función de la longitud y complejidad de la obra, éstos deberían contarse a partir de la fecha de inicio de las obras de puesta a punto, es decir a los 10 o 15 días de aprobado el Diseño Ejecutivo, y el plazo de considerar los tiempos de movilización + replanteo + tiempo necesario para ejecutar las obras de puesta a punto. Debe emitirse la orden de inicio una vez aprobado el Diseño Ejecutivo presentado por la

Contratista y estudiado por la fiscalización. A su vez, iniciada las obras de puesta a punto, regirán los plazos para la puesta a punto, en una primera etapa se deben lograr los estándares básicos y en la segunda que se produce a continuación de la etapa 1 y finaliza a la terminación del plazo de las obras de puesta a punto.

- Se debería definir una metodología más precisa para implementación de las multas y bonificaciones, ya que pueden quedar a criterio de la fiscalizadora. Asimismo la severidad de la penalidad y la extensión del incumplimiento. Además, se recomienda aplicar penalidades ante un nivel de servicio determinado (ejemplo 95%) y no aplicar bonificaciones, ya que conceptualmente el contratista tiene que cumplir ese umbral mínimo y no debería recibir bonificación por un umbral que debe cumplir si lo supera.

- Otras Intervenciones Obligatorias: En el DDL podrían incorporarse otras actividades tales como: control de cargas, inventario vial, relevamiento de estado, medición de rugosidades, mediciones de tránsito.

- 2. Microempresas:

    - El pago a las microempresas debería guardar relación con la producción obtenida, re analizando la modalidad de Contratación de las mismas.
    - Analizar la forma de contratar a las Microempresas: permitir que sea la Contratista la responsable de elegir las personas con quienes deberá trabajar.
    - Permitir el uso de máquinas.
    - Establecer horarios de trabajo para las microempresas. Considerar la posibilidad de aplicación de multas para las microempresas en caso de no cumplir con sus estándares

- 3. Fiscalizadoras:

    - Prever aplicación de multas a la Fiscalización por incumplimientos.
    - Pago a la Fiscalización por mes-hombre y gastos reembolsables.
    - Incluir en el contrato de la fiscalización cláusulas para exigir la realización de los trabajos en forma y plazo.
    - Incorporar la medición de reflectancia en la señalización vertical e incluir en Contrato de Fiscalización, incluyendo los umbrales en los estándares de cumplimiento.

4. Respecto al Mejoramiento de Caminos No Pavimentados:

Este proyecto previó en un principio el mejoramiento de 960 km de Caminos No Pavimentados. En el 2008 se solicitó la inclusión de la Ruta IV (San Ignacio – Pilar) en los contratos GMANS. Esto requirió la asignación de importantes recursos del Préstamo, lo cual obligó a disminuir los Caminos No Pavimentados a ser mejorados dentro del Proyecto.

Los tramos que finalmente fueron ejecutados suman 140 km de longitud y se detallan a continuación:

- **San Pedro**:
  - Lote 1:Chore - Colonia Cocuerá - San Pablo
  - Lote 2: Chore - Jhugua Rey
  - Lote 3: Jhugua Rey - Gral. Aquino

- **Caaguazú**: Colonias Unidas - Yjhovy Central - Raúl A. Oviedo - J. Eulogio Estigarribia

- **Caazapá**: Caazapá – Boquerón

Respecto a los tramos en el Departamento de **San Pedro**, los Lote 2 y 3 se ejecutaron sin mayores problemas, en el plazo establecido en el contrato. Respecto al Lote 1; se generaron disturbios por parte de los pobladores, quienes se encontraron divididos en 2 grupos en la zona de Colonia Cocuerá: uno que pedía que el mejoramiento pase por la zona conocida como Tapé Tuja, y otro que pedía que pase por el cementerio. Amenazaron con no permitir que los equipo trabajen si no se cumplía con sus solicitudes, incluso con quemar los mismos. Se solicitó al Banco Mundial la No Objeción para ampliar el contrato y mejorar ambos tramos, para evitar el problema social generado, y se procedió a la elaboración del diseño del tramo faltante (4 km aprox.), posteriormente se solicitó la cotización de los adicionales generados y se procedió a ejecutar dichos adicionales. Todo este proceso generó también prórrogas en los plazos.

Lección aprendida: los problemas sociales pueden ocasionar problemas graves en las Obras como retrasos, perjuicios a los equipos del contratista, sobrecostos, tanto para el contratista como para la fiscalización.

Recomendación: una mejor evaluación de los grupos sociales, y socialización del proyecto en etapas previas al llamado.

Caaguazú: en el diseño no se previeron las cañerías de la red de agua. Esto generó atrasos en los inicios de la Obra. Se solicitó el levantamiento de red existente y el re diseño de la nueva red. Esto generó prorroga de plazos y adicionales.

Recomendación: Minimizar los imprevistos, incluyendo el relevamiento de las redes subterráneas existentes en los pliegos de Diseño.

Caazapá: Esta obra se encuentra en zonas de prolongadas inundaciones, además el plano de fundación de los pilotes para puentes resultó estar en un estrato más profundo de lo previsto. Esto generó prórrogas muy importantes, con un plazo final mayor al triple del plazo inicial.

Recomendación: Incluir verificación de los estudios de suelo del periodo de diseño entre las actividades del Contratista, a fin de verificar la profundidad de las fundaciones. Además analizar mejor las zonas inundables, y prever plazos más realistas según la zona.

Annex 8. Comments of Cofinanciers and Other Partners/Stakeholders

n.a

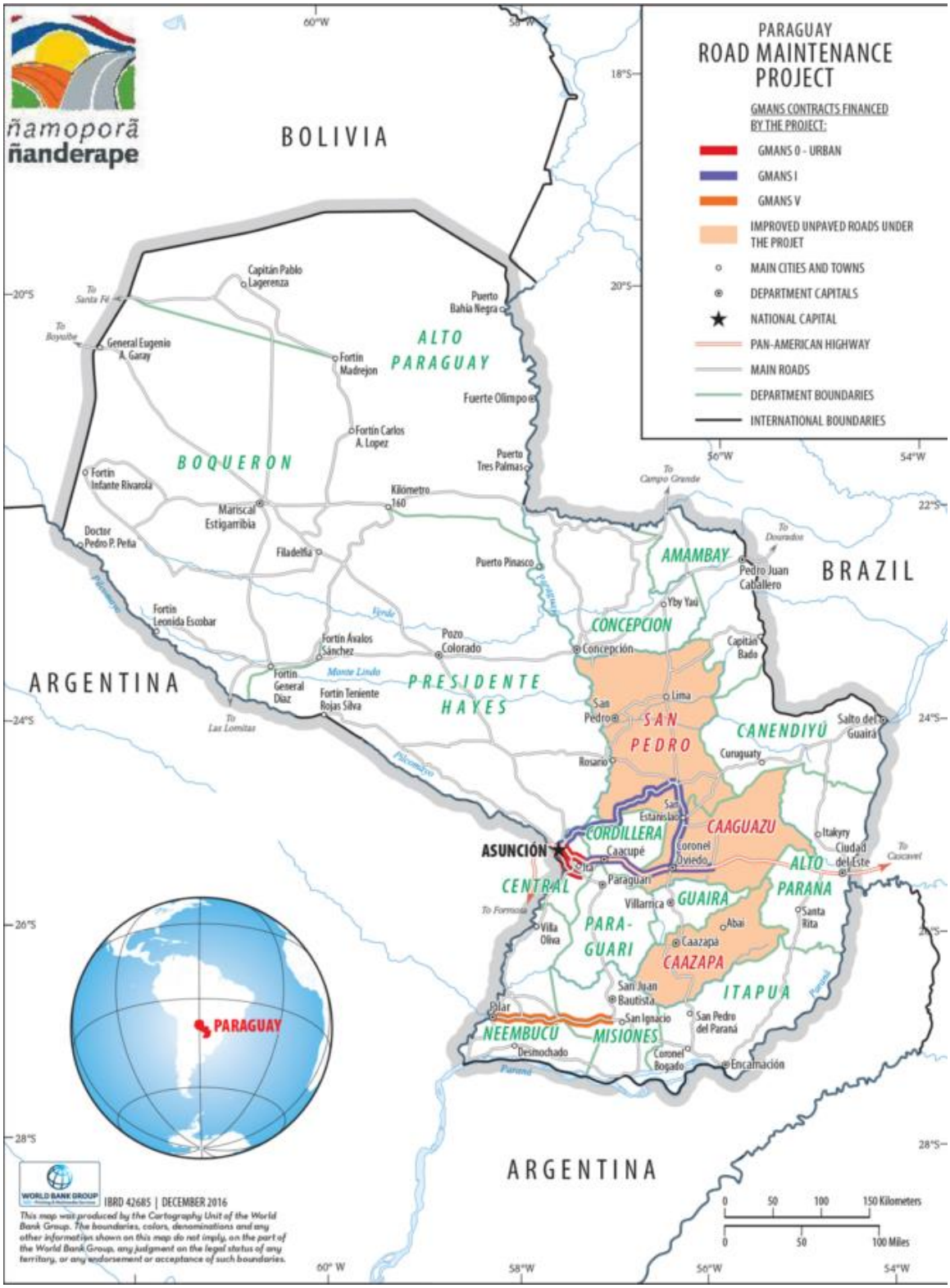
Annex 9. List of Supporting Documents

Project Appraisal Document
Project Loan Agreement
Project Restructuring Papers
Project Preparation, Appraisal and Supervision Aide Memoires
Implementation Status Reports

Operational Plan for the 3 Community Centers *Plan de Uso de Centros Comuntarios en los Tres Departamentos de San Pedro, Caazapa y Caaguazu,* Instituto Paraguayo del Indigena, June 2016

Summary of Contracts and Final Costs *Cuadro de Contratos,* June 2016
Traffic Estimates 2011-2015 by toll station
Update Vehicle Operating Cost Estimates 2016

MAP


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PARAGUAY
ROAD MAINTENANCE
PROJECT
GMANS CONTRACTS FINANCED BY THE PROJECT:
GMANS 0 - URBAN
GMANS I
GMANS V
IMPROVED UNPAVED ROADS UNDER THE PROJET
MAIN CITIES AND TOWNS
DEPARTMENT CAPITALS
NATIONAL CAPITAL
PAN-AMERICAN HIGHWAY
MAIN ROADS
DEPARTMENT BOUNDARIES
INTERNATIONAL BOUNDARIES
BOLIVIA
BRAZIL
ARGENTINA
ARGENTINA
ALTO PARAGUAY
BOQUERON
PRESIDENTE HAYES
AMAMBAY
CONCEPCION
SAN PEDRO
CANENDIYU
CORDILLERA
CAAGUAZU
ASUNCIÓN
CENTRAL
ALTO PARANA
GUAIRA
PARA-GUARI
CAAZAPA
ITAPUA
NEEMBUCU
MISIONES
Capitán Pablo Lagerenza
Puerto Bahía Negra
General Eugenio A. Garay
Fortin Madrejon
Fuerte Olimpo
Fortin Carlos A. Lopez
Puerto Tres Palmas
Fortin Infante Rivarola
Mariscal Estigarribia
Kilometro 160
Doctor Pedro P. Peña
Filadelfia
Puerto Pinasco
Fortin Leonida Escobar
Fortin Ávalos Sánchez
Pozo Colorado
Fortin General Diaz
Fortin Teniente Rojas Silva
Pedro Juan Caballero
Yby Yau
Concepción
Capitán Bado
Lima
San Pedro
Salto del Guairá
Curuguaty
Rosario
San Estanislao
Coronel Oviedo
Caacupé
Itakyry
Ciudad del Este
Paraguarí
Villarrica
Santa Rita
Abaí
Villa Oliva
Caazapá
San Juan Bautista
Pilar
San Ignacio
San Pedro del Paraná
Desmochado
Coronel Bogado
Encarnación
PARAGUAY
WORLD BANK GROUP
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0 50 100 150 Kilometers
0 50 100 Miles
60°W
58°W
56°W
54°W
18°S
20°S
22°S
24°S
26°S
28°S